RERIS

Robot Interface System

USER & ADMINISTRATOR MANUAL

Version 1.0

Developed by

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1. Introduction

RERIS (Robot Interface System) is a web-based graphical user interface designed to simplify the control, mapping, and autonomous navigation of mobile robots running on the ROS 2 framework. Operators can access the system from any modern web browser on the same local network as the robot — no additional software installation is required on the client device.

This manual is structured as follows:

- Part A — Administrator Guide (Chapters 2–5): installation, setup, user management, topic configuration, and technical reference. Complete all steps before handing the system to regular users.
- Part B — User Guide (Chapters 6–14): daily operation, dashboard navigation, map recording, toolbox reference, tool inspection, and map export.

1.1 System Overview

Product Name	RERIS — Robot Interface System
Version	1.0.0
Developed by	Mr. Napat Chanlarm & Mr. Pathanadet Srianan
Department	Mechatronics Engineering
Institution	College of Industrial Technology, King Mongkut's University of Technology North Bangkok
Frontend	Vue 3 + Quasar Framework (Web Browser)
Backend	C# ASP.NET Core 8.0 REST API
Database	MongoDB 7.0
Robot Operating System	ROS 2 Humble
Communication	WebRTC Peer-to-Peer Data Channel
Default Port	9000 (HTTP)

1.2 Client Requirements

Web Browser	Chrome 110+, Firefox 110+, Edge 110+, Safari 16+
Network	Must be on the same local network as the robot
Internet Access	Not required — fully offline capable
Screen Resolution	1280 x 720 minimum recommended

PART A

Administrator Guide

2. Installation

This chapter must be completed by the Administrator before any user can access the system. All steps are performed on the robot over SSH.

2.1 Prerequisites

Operating System	Ubuntu 22.04 LTS on the robot
ROS 2	Humble Hawksbill — installed and sourced
Network	Robot reachable via SSH from the install machine
Disk Space	Minimum 4 GB free on the robot
Privileges	sudo access required on the robot

2.2 ROS 2 Dependencies

In addition to a base ROS 2 Humble installation, the following packages must be installed on the robot before running the RERIS installer:

slam_toolbox

slam_toolbox provides the Simultaneous Localisation and Mapping (SLAM) capability. RERIS relies on slam_toolbox to build and update the 2D OccupancyGrid map in real time during the recording phase.

```
sudo apt install ros-humble-slam-toolbox
```

Nav2 Navigation Stack

Nav2 is the autonomous navigation stack used by RERIS to plan paths and execute movement commands. The following core Nav2 packages are required:

```
sudo apt install ros-humble-navigation2
sudo apt install ros-humble-nav2-bringup
```

Hardware Driver Package (Required — Not Included)

RERIS sends velocity commands to the robot by publishing to the /cmd_vel topic. The robot must have its own hardware driver package that subscribes to this topic and forwards the linear velocity (Vx) and lateral velocity (Vy) values to the wheel controller. The hardware package must also publish the following data for RERIS to function correctly:

Odometry	Wheel encoder data published to /odom — provides robot position (x, y) and heading (yaw) via dead-reckoning.
Wheel Speed	Individual wheel speed feedback used by the odometry estimator.
IMU Data	Inertial Measurement Unit data published to /imu — fused with odometry by slam_toolbox for improved localisation accuracy.

NOTE RERIS does not include or provide the hardware driver package. The integrator is responsible for implementing and launching the driver for the specific robot platform. slam_toolbox requires a valid /odom and /scan (LiDAR) topic to build maps.

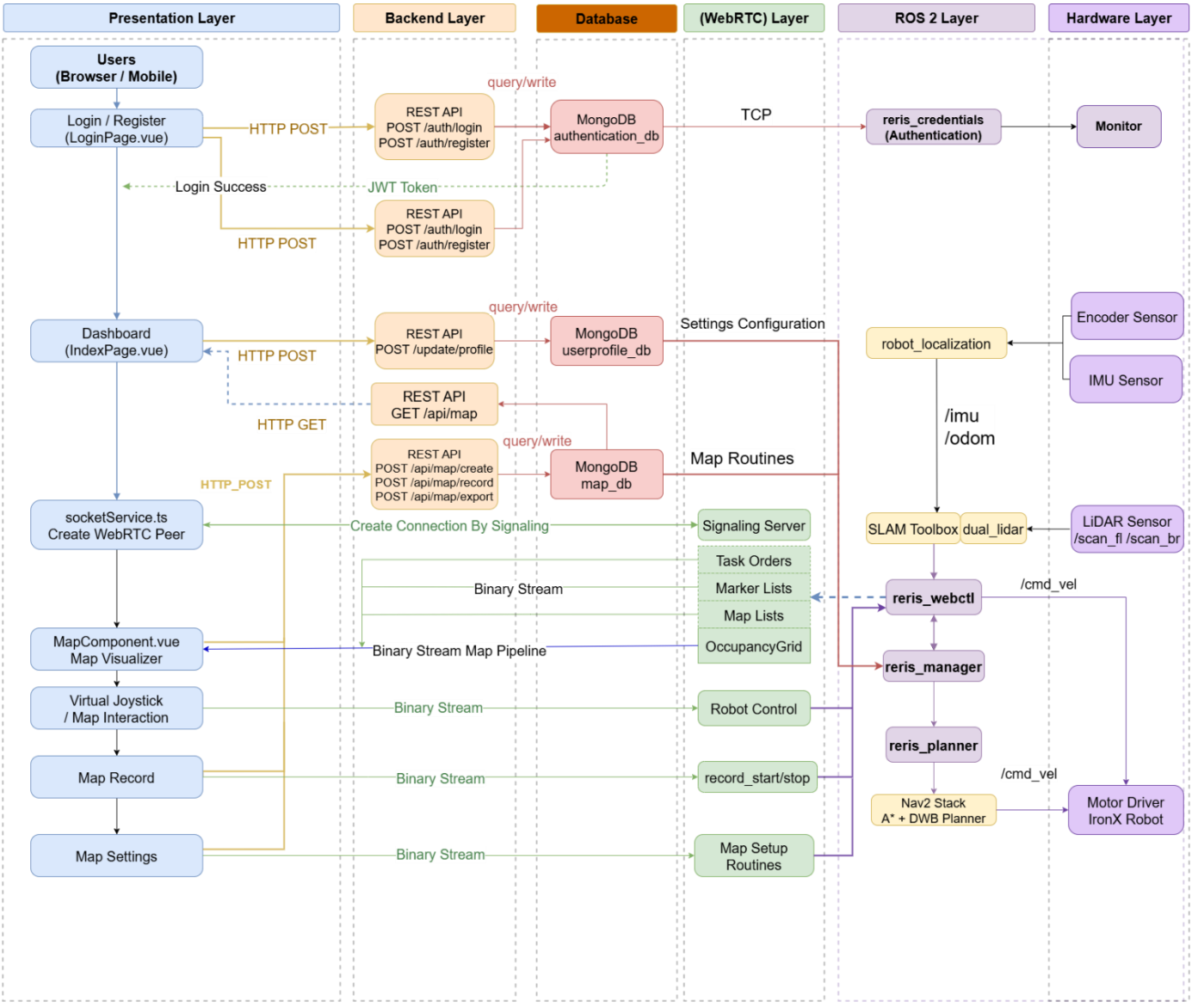


Figure 2-1: ROS 2 topic flow diagram — /cmd_vel, /odom, /scan, /imu connections between RERIS and robot hardware

2.3 Connecting to the Robot via SSH

From a terminal on your computer, connect to the robot:

```
ssh <username>@<Robot IP address>
```

2.4 Preparing the Workspace

Copy the reris_controller package to the robot. The directory structure must be as follows:

```
~/reris_ws/
    src/
        reris_controller/
            reris_install.sh
            launch/
                reris_deployed.launch.py
            reris_controller/
            package.xml
            setup.py
```

2.5 Running the Installer

Navigate into the src directory and execute the installer:

```
cd ~/reris_ws/src
chmod +x reris_install.sh
./reris_install.sh
```

The installer performs the following steps automatically:

1. Updates system package lists.
2. Installs MongoDB 7.0 and starts the mongod service.
3. Installs .NET 8.0 Runtime and ASP.NET Core 8.0 Runtime.
4. Builds the ROS 2 workspace using colcon build --symlink-install.
5. Appends workspace source lines to ~/.bashrc.
6. Prompts for the startup mode selection (see Section 2.6).

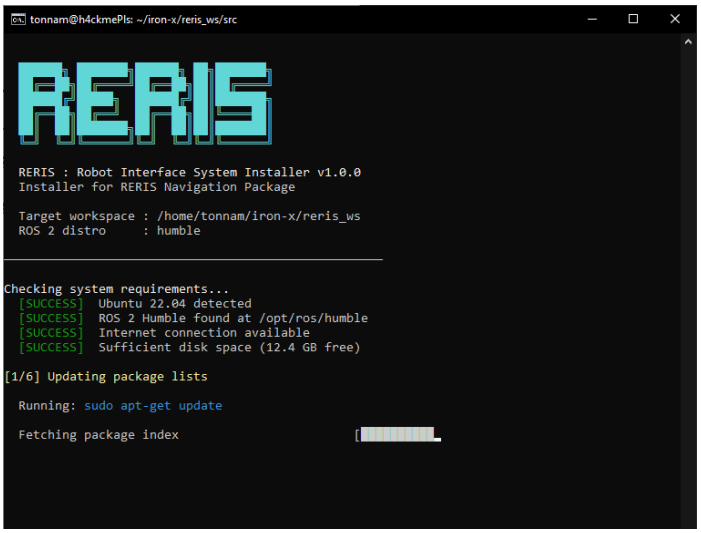


Figure 2-2-1: Terminal output during reris_install.sh execution showing each installation step

2.6 Selecting the Startup Mode

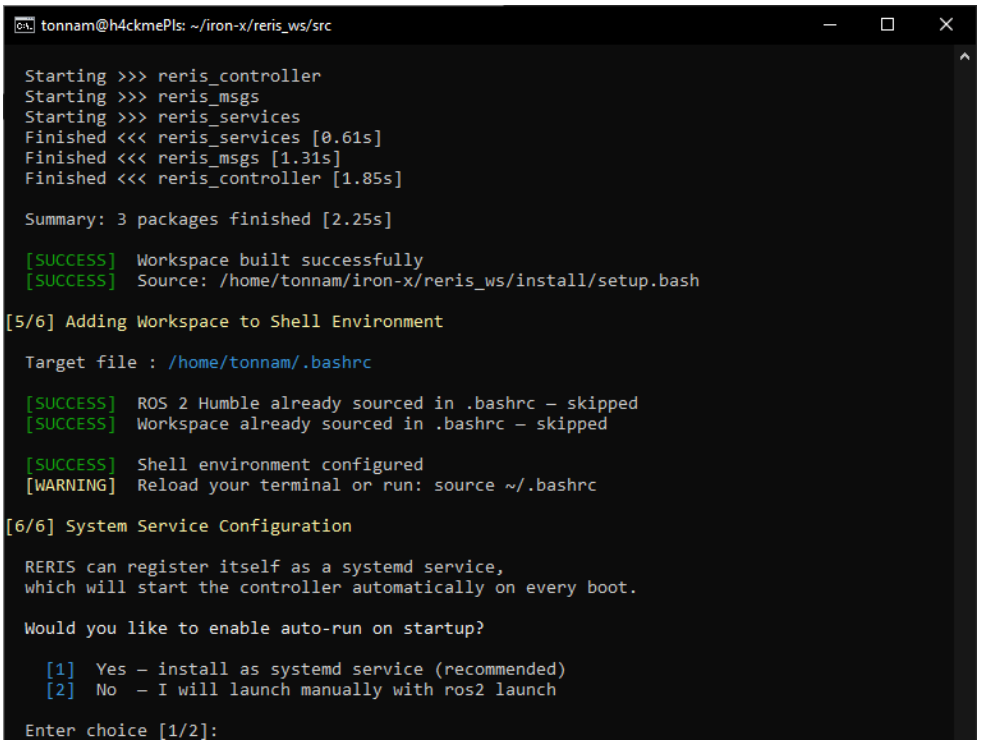


Figure 2-2-2: Terminal output during reris_install.sh execution showing ask about mode

Mode 1 — Auto-start	Registers RERIS as a systemd service. Launches automatically on every boot. Recommended for production.
Mode 2 — Manual-start	No service registered. Must be started manually each session.

2.7 Launching the System Manually (Mode 2)

```
source /opt/ros/humble/setup.bash
source ~/reris_ws/install/setup.bash
ros2 launch reris_controller reris_deployed.launch.py
```

The terminal prints the access URL once all nodes are ready:

```
http://<Robot IP address>:9000
```

2.8 Managing the Auto-start Service (Mode 1)

Check status	sudo systemctl status reris_deployed
Restart	sudo systemctl restart reris_deployed
Stop	sudo systemctl stop reris_deployed
Disable on boot	sudo systemctl disable reris_deployed

3. First-Time Setup — Administrator Account

After the system is running, the Administrator must register the first account before any user can access the interface. RERIS strictly enforces that the Administrator account is registered before any regular User account.

> **WARNING** The very first account registered must have the Administrator role. If a User account is created first by mistake, it must be deleted and the process restarted.

3.1 Creating the Administrator Account

7. Open a browser and navigate to http://<Robot IP address>:9000.
8. Click Register on the Login page.
9. Fill in the registration form:

Username	Choose a unique administrator username (no spaces)
Password	Minimum 8 characters
Confirm Password	Must match the password field
Role	Select Administrator

10. Click Register.
11. The system generates a Security Code — save it immediately. It is shown only once and is required for password reset.

> **NOTE** RERIS runs on a closed local network without internet. The Security Code is the offline identity verification mechanism. If lost, the Administrator account cannot be recovered without direct database access.

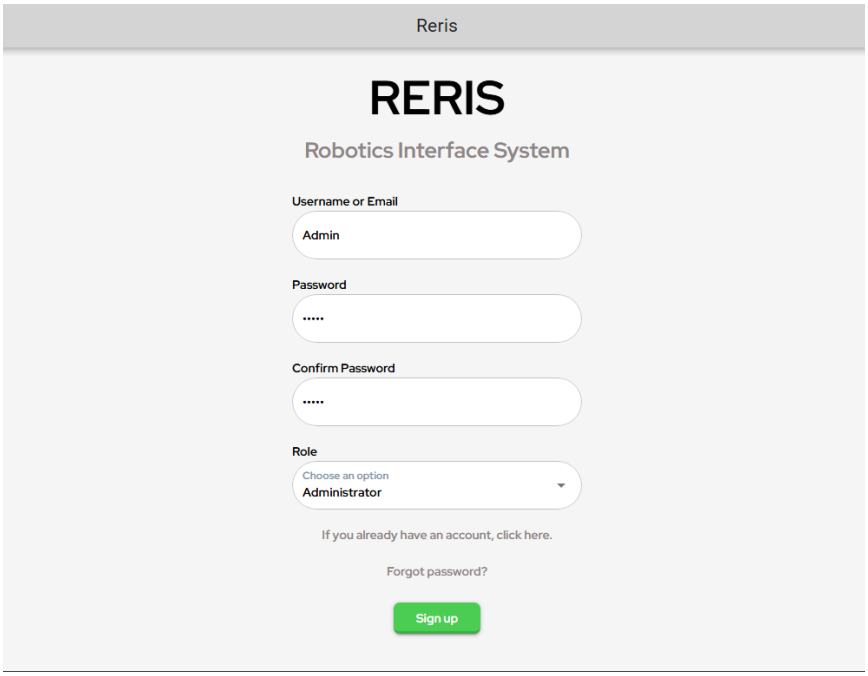


Figure 3-1-1: Register page showing the role selection

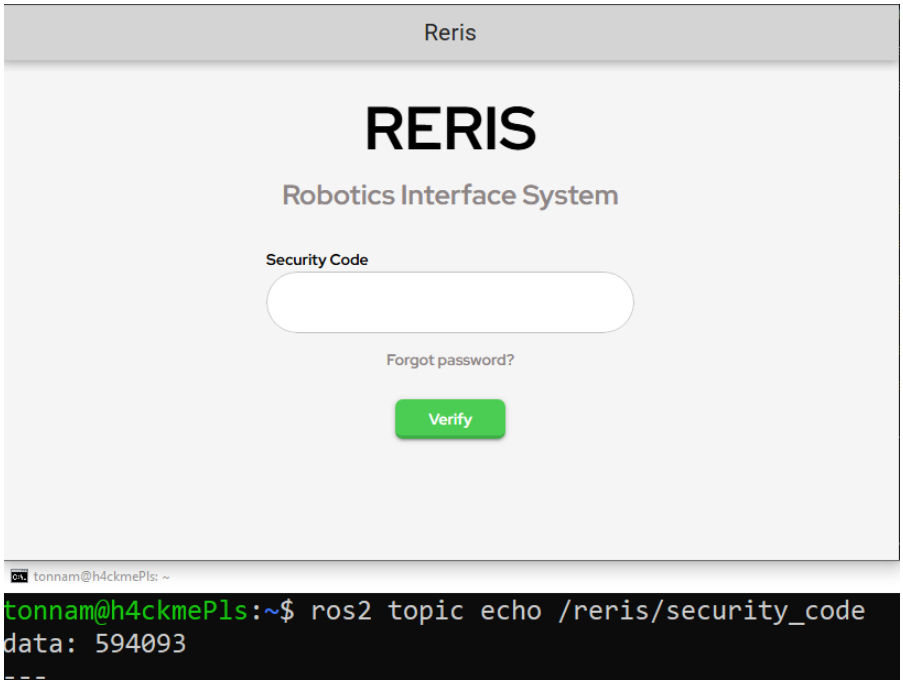


Figure 3-1-2: Security Code generation field

After setting your password during account registration (for any role), you must verify your security code. This code is published by the reris_webctl node on the /reris/security_code topic. You can integrate this into your system to retrieve and display the code on any device.

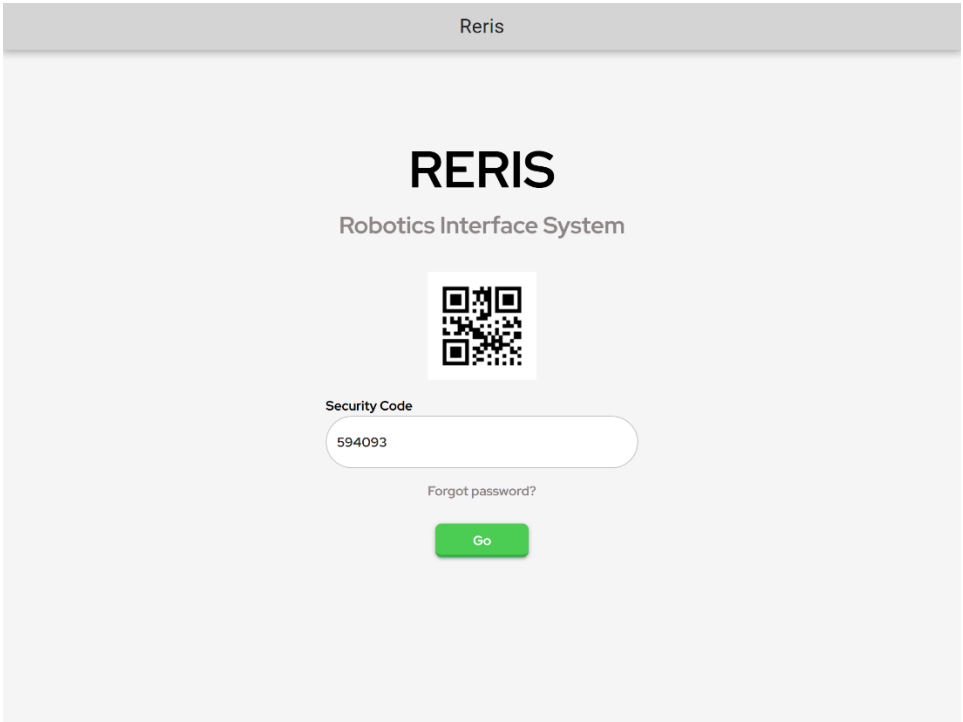


Figure 3-1-3: QR Code Security Code

You must remember this security code or capture the QR Code because that use when you forget password

3.2 Logging In as Administrator

12. Navigate to http://<Robot IP address>:9000.
13. Enter your Username and Password and click Sign In.
14. You will be directed to the Dashboard.

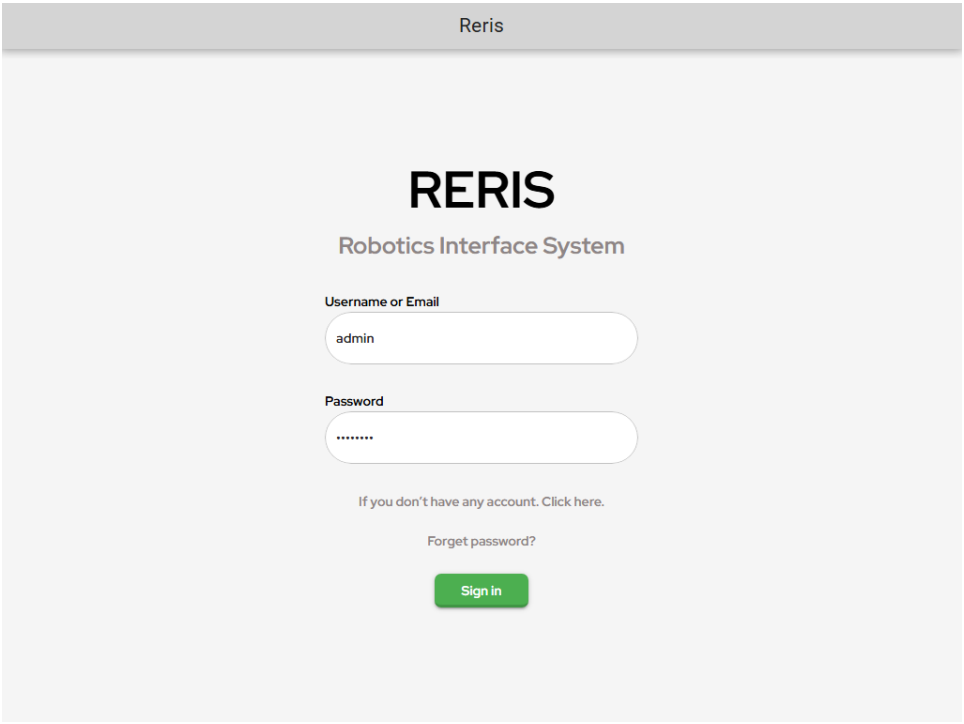


Figure 3-2-1: Login page

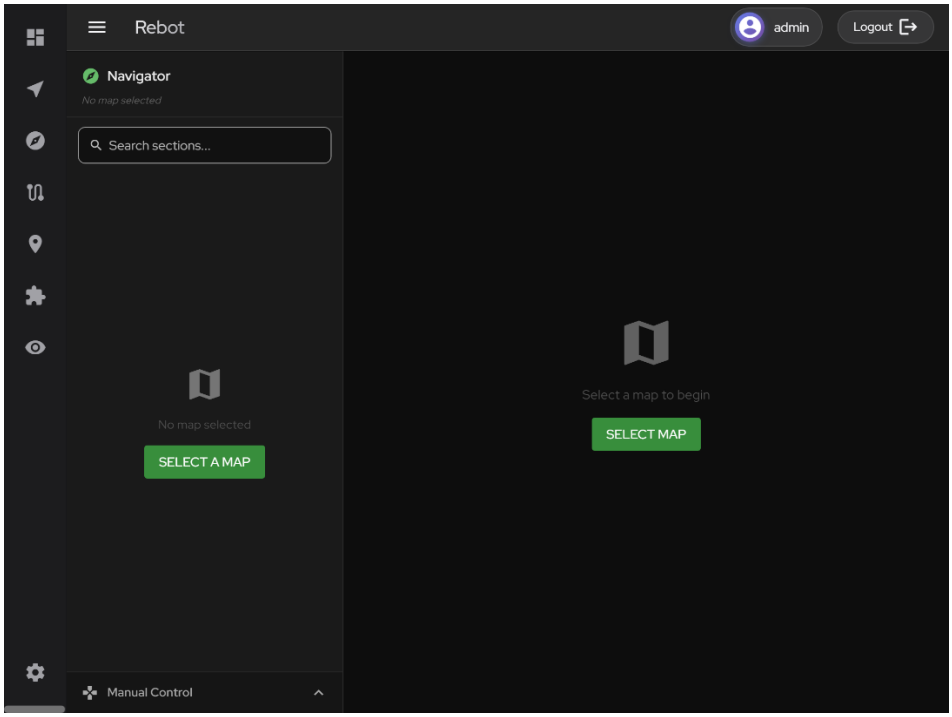


Figure 3-2-2: Dashboard page

4. User & Role Management

All user management is performed by the Administrator through the Profile page.

4.1 Role Management

Navigate to Profile → Role Management to view all registered users, their roles, and account creation dates.

- Click Modify next to a user to change their role between Administrator and User.
- The first Administrator is created at registration. Additional Administrators must be promoted here.

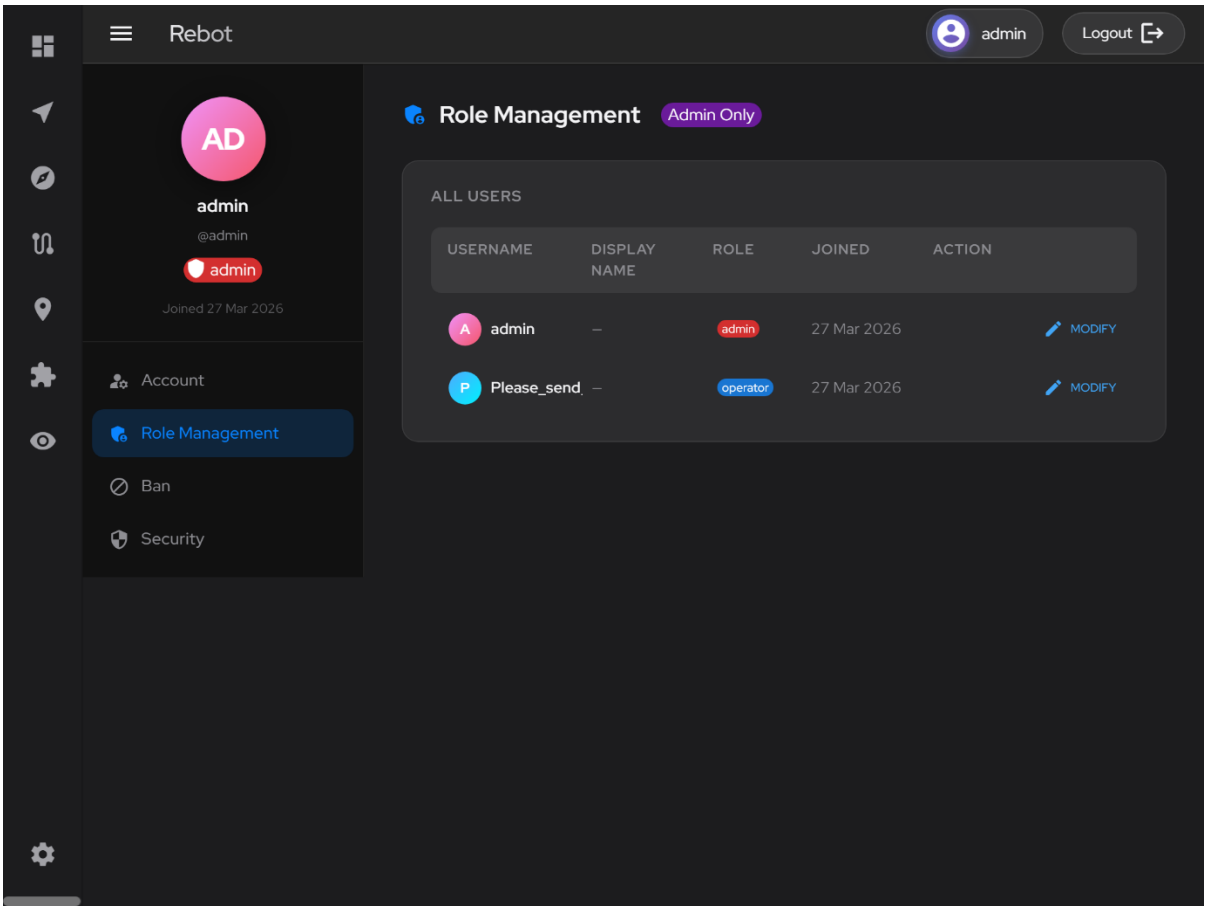


Figure 4-1-1: Role Management page showing the user list and Modify button

4.2 Approving New User Accounts

Users can self-register using the User role. The Administrator reviews all accounts in Role Management and can modify roles or ban accounts as needed.

4.3 Ban Management

Navigate to Profile → Ban to suspend a user account. Banned users cannot login until the ban is lifted.

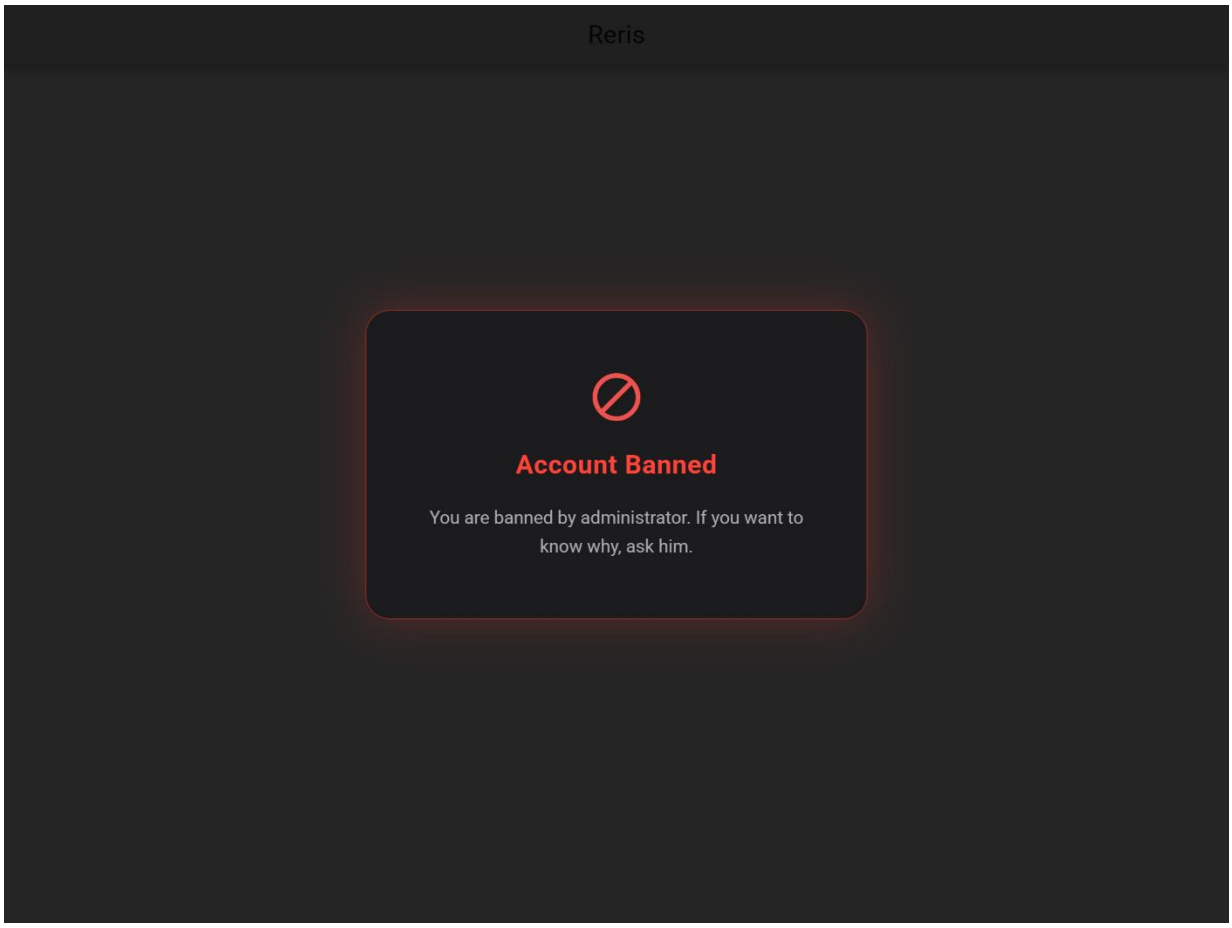


Figure 4-1-2: Ban Mangement

4.4 Security — Regenerating Security Codes

Navigate to Profile → Security to generate a new Security Code for a managed account when a user has lost their code and cannot reset their own password.

5. System Settings & Technical Reference

Navigate to Settings in the left navigation bar. This page is restricted to Administrator accounts.

5.1 Topic & Connection Configuration

RERIS allows the Administrator to configure all ROS 2 topic names and connection parameters directly from the web interface under Settings. No code changes or file edits on the robot are required. The following parameters are configurable:

Robot IP Address	IP address of the robot on the local network (default: 192.168.10.2)
API Port	HTTP port for the REST API backend (default: 5184)
Telemetry Port	WebRTC/WebSocket channel 1 — Odometry, node status, plan data (default: 8440)
Map Port	WebRTC/WebSocket channel 2 — OccupancyGrid map data (default: 8441)
/cmd_vel Topic	Topic name that RERIS publishes velocity commands to (Vx, Vy). Default: /cmd_vel
/odom Topic	Topic name for robot odometry (position + orientation as Quaternion). Default: /odom
/map Topic	Topic name for the OccupancyGrid published by slam_toolbox. Default: /map
/scan Topic	Topic name for LiDAR scan data consumed by slam_toolbox. Default: /scan
/imu Topic	Topic name for IMU data used in odometry fusion. Default: /imu

NOTE Changes to topic names take effect on the next RERIS launch. Ensure the hardware driver package publishes to matching topic names.

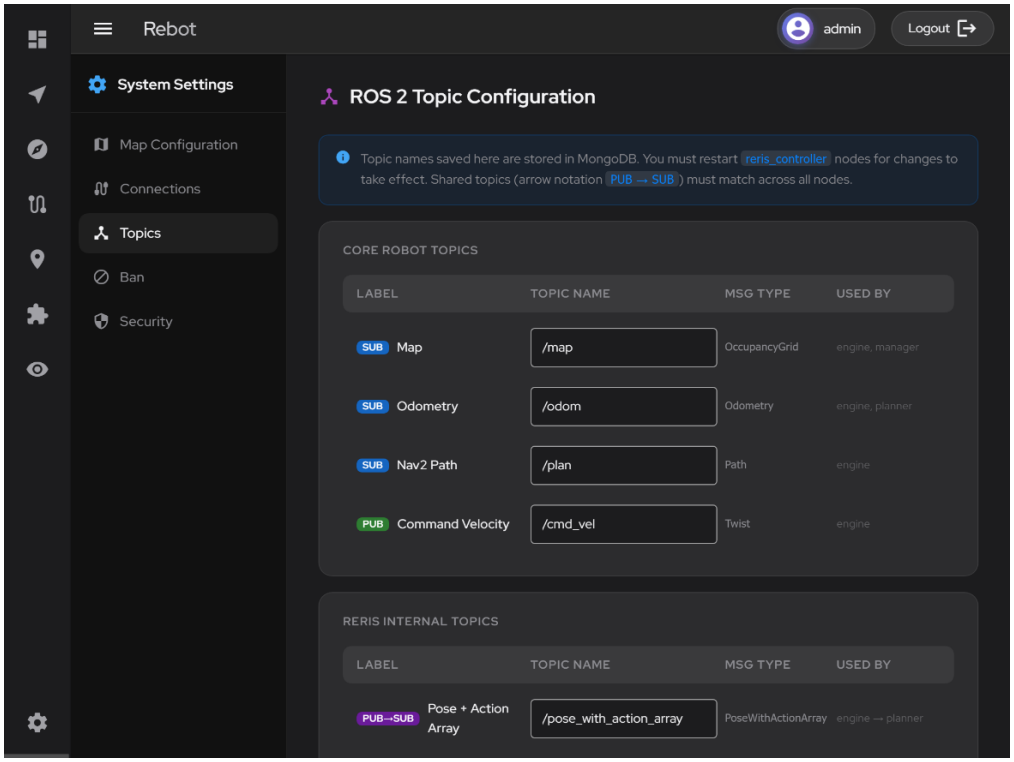


Figure 5-1: Settings page — Connection and Topic Configuration panel

5.2 Map Configuration

Configure global map parameters such as default resolution, map save path, and slam_toolbox mode (online async / lifelong).

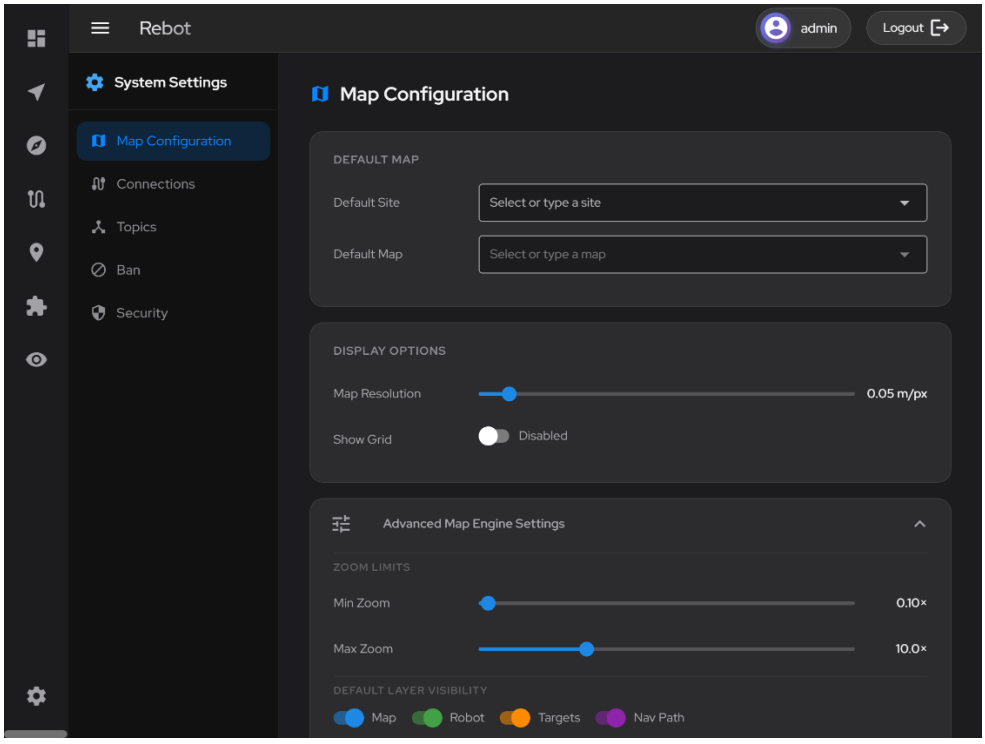


Figure 5-2: Settings page — Map Configuration panel

5.3 Active Connections

View all active WebRTC peer-to-peer connections between client browsers and the robot backend. Use this panel to diagnose communication issues or confirm data channels are established.

5.4 Ban (System-level)

Block specific IP addresses or device identifiers from connecting to the RERIS interface. Separate from account-level ban in Profile → Ban.

5.5 System Architecture

Presentation Layer	Vue 3 + Quasar Framework in the browser. All UI rendering and WebRTC client logic.
WebRTC Layer	Peer-to-peer channels carrying telemetry, map data, and video/PointCloud streams.
Backend Layer	C# ASP.NET Core 8.0 REST API — authentication (JWT), map management, user administration.
Database Layer	MongoDB 7.0 — three independent databases for auth, profiles, and maps.
ROS 2 Layer	ROS 2 Humble nodes — SLAM, Nav2, sensor drivers, hardware control.

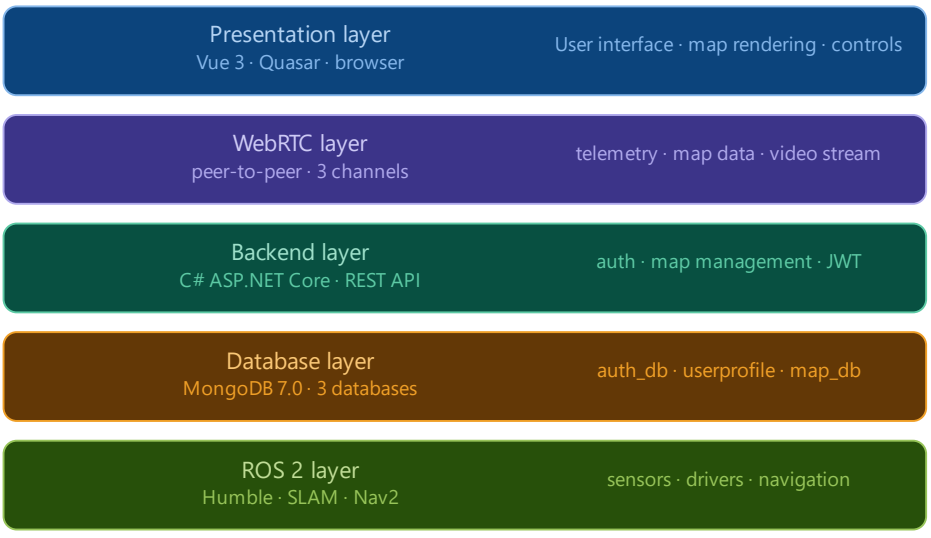


Figure 5-3: RERIS system architecture diagram showing all five layers

5.6 WebRTC Channels

Channel 1 — Telemetry	Odometry, node status, navigation plan. Port 8440. Real-time robot tracking.
Channel 2 — Map Data	OccupancyGrid updates from slam_toolbox. Port 8441. Separated to avoid bandwidth contention.
Channel 3 — Video	Reserved for image streams and LiDAR PointCloud. Designed for future extension.

5.7 ROS 2 Package Structure

Package	reris_controller
reris_webctl	Web control bridge — manages WebRTC signalling and routing between browser and ROS 2 topics.
reris_planner	Navigation planner — interfaces with Nav2 for goal commands and multi-waypoint routing.
reris_manager	System resource manager — manages node lifecycle, map export/import, and health monitoring.
reris_analyze	Analysis module — reserved for telemetry logging and analytics (future extension).

5.8 REST API Reference

All endpoints except /auth/* require a JWT Bearer token in the Authorization header.

Method	Endpoint	Purpose
POST	/auth/login	Authenticate with Username and Password. Returns a JWT token.
POST	/auth/register	Create a new user account and generate a Security Code.
POST	/auth/secure/code	Generate a new Security Code for an existing user.
POST	/auth/forget	Reset password by verifying the Security Code.
POST	/update/profile	Update the authenticated user's profile.
GET	/api/map	Retrieve all maps belonging to the authenticated user.
POST	/api/map/create	Create a new map entry under the specified Site.
POST	/api/map/record	Start a SLAM recording session.
POST	/api/map/export	Export the recorded map and save Zones/Markers to MongoDB.
PATCH	/api/map/update	Update map status or metadata.
POST	/api/map/remove	Delete a map from its Site.

5.9 Database Reference

authentication_db — Users	Username, hashed Password, JWT tokens, Security Code.
userprofile — Profiles	User settings, role (Administrator / User), ban status.
map_db — Maps	Site names and Map arrays: name, author, status, file location, Zones, Markers.

5.10 Troubleshooting

Cannot access web interface	Check: sudo systemctl status reris_deployed. Verify robot is on the network.
Map not updating	Check Settings → Connections. Reload browser to re-establish WebRTC connection.
colcon build fails	Source ROS 2 first: source /opt/ros/humble/setup.bash
MongoDB connection error	sudo systemctl restart mongod
slam_toolbox not mapping	Verify /scan topic is publishing. Check topic name in Settings.
Robot not moving after goal	Confirm reris_planner is running: ros2 node list. Check /cmd_vel topic name in Settings.
Forgot Security Code	Administrator generates new code via Profile → Security.

PART B

User Guide

6. Getting Started

Before proceeding, confirm with the Administrator that installation is complete and the system is running.

6.1 Accessing the System

Open a web browser on any device on the same network as the robot:

```
http://<Robot IP address>:9000
```

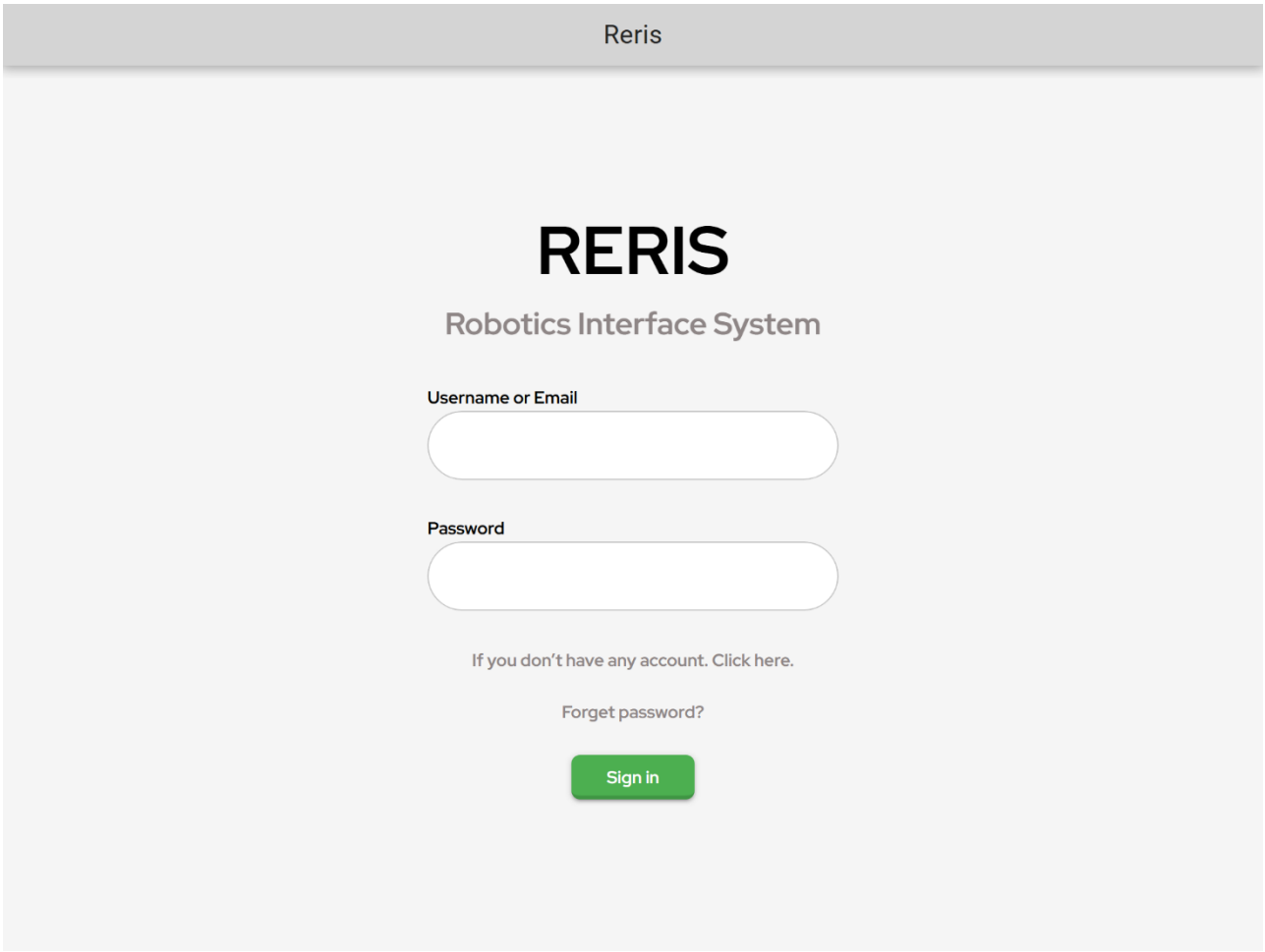


Figure 6-1: RERIS web interface home screen as seen in the browser

6.2 User Viewer & Operator / Registration

Click Register on the Login page and fill in:

Username	Unique username (no spaces)
Password	Minimum 8 characters
Confirm Password	Must match
Role	Select Viewer / Operator

After setting your password during account registration (for any role), you must verify your security code. This code is published by the reris_webctl node on the /reris/security_code topic. You can integrate this into your system to retrieve and display the code on any device.

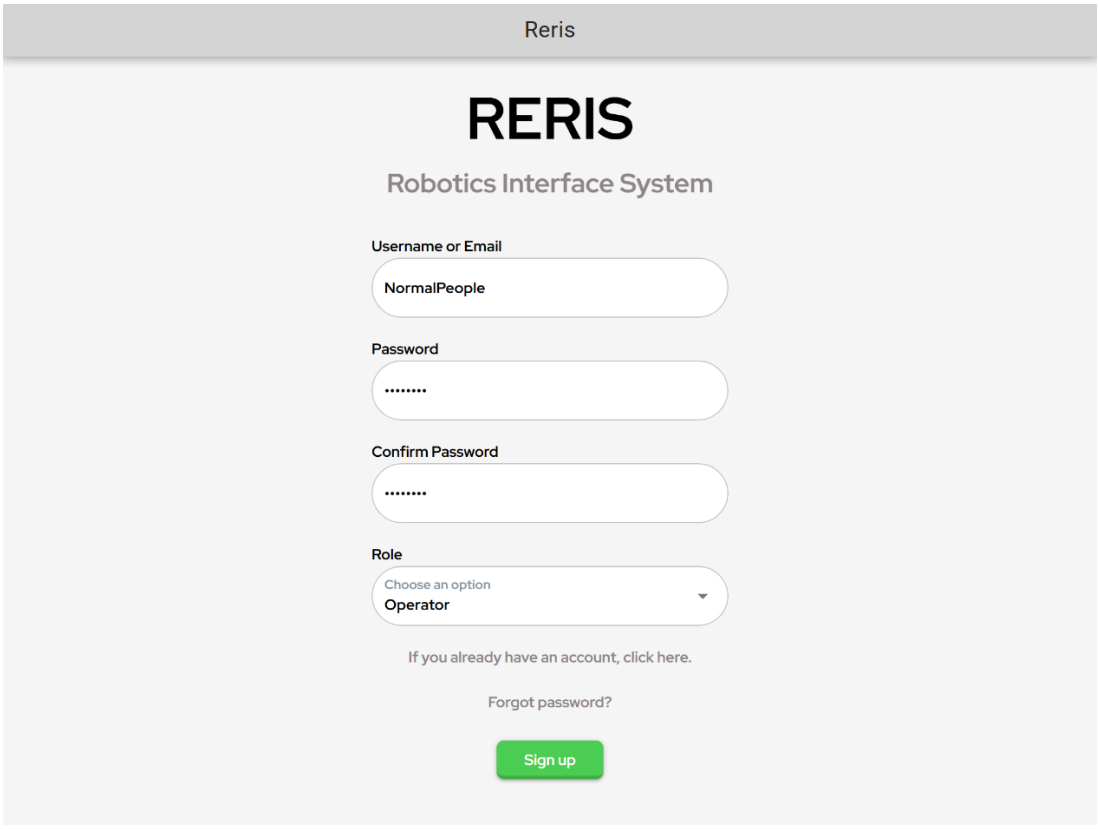


Figure 6-2: Register page

6.3 Login

Enter Username and Password and click Sign In. You will be directed to the Dashboard.

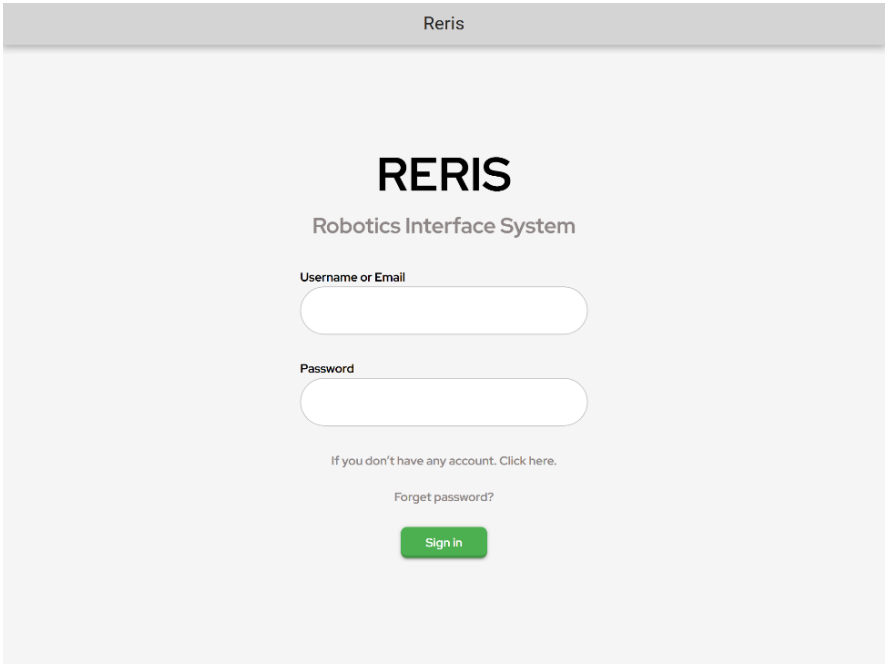


Figure 6-3: Login page

6.4 Password Reset

15. Click Forgot Password on the Login page.
16. Enter your Username and Security Code saved at registration.
17. Enter and confirm your new password, then click Confirm.

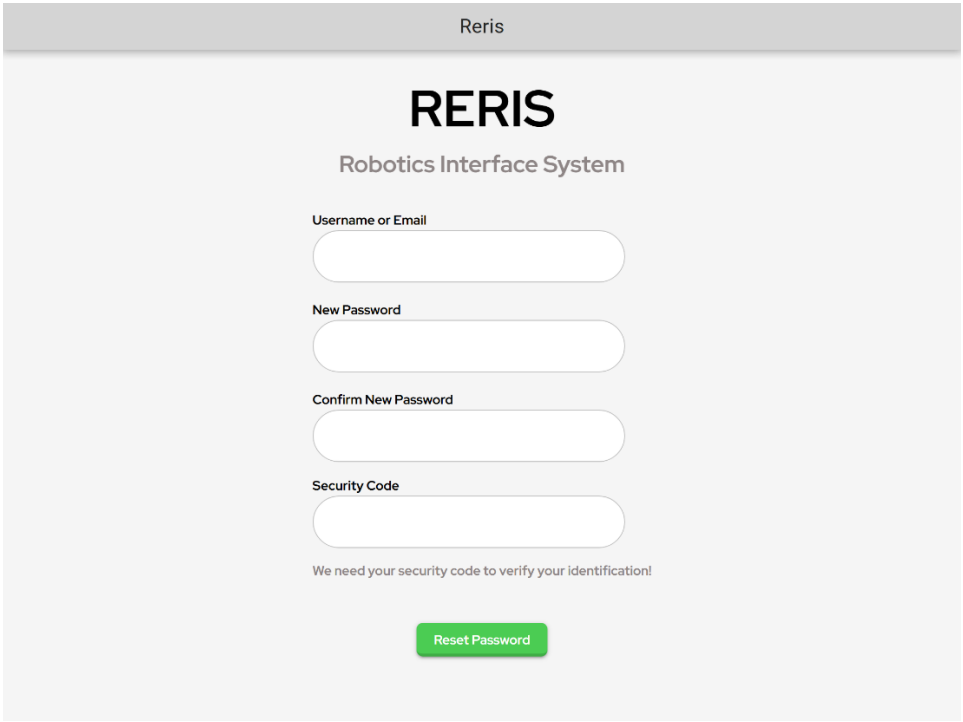


Figure 6-4: Forgot Password page

7. Dashboard

The Dashboard is the primary screen for monitoring and controlling the robot in real time.

7.1 Map View

The map panel renders a live 2D occupancy map. The robot is shown as a green triangle with an arrow indicating heading direction. Saved Zones and Markers from previous mapping sessions are overlaid on the map.

Green triangle / arrow	Robot's current position and heading direction
Coloured rectangles	Navigation zones defined during mapping
Pins with arrows	Navigation markers (waypoints) with target headings
Red rectangles	Restricted zones — robot will not enter

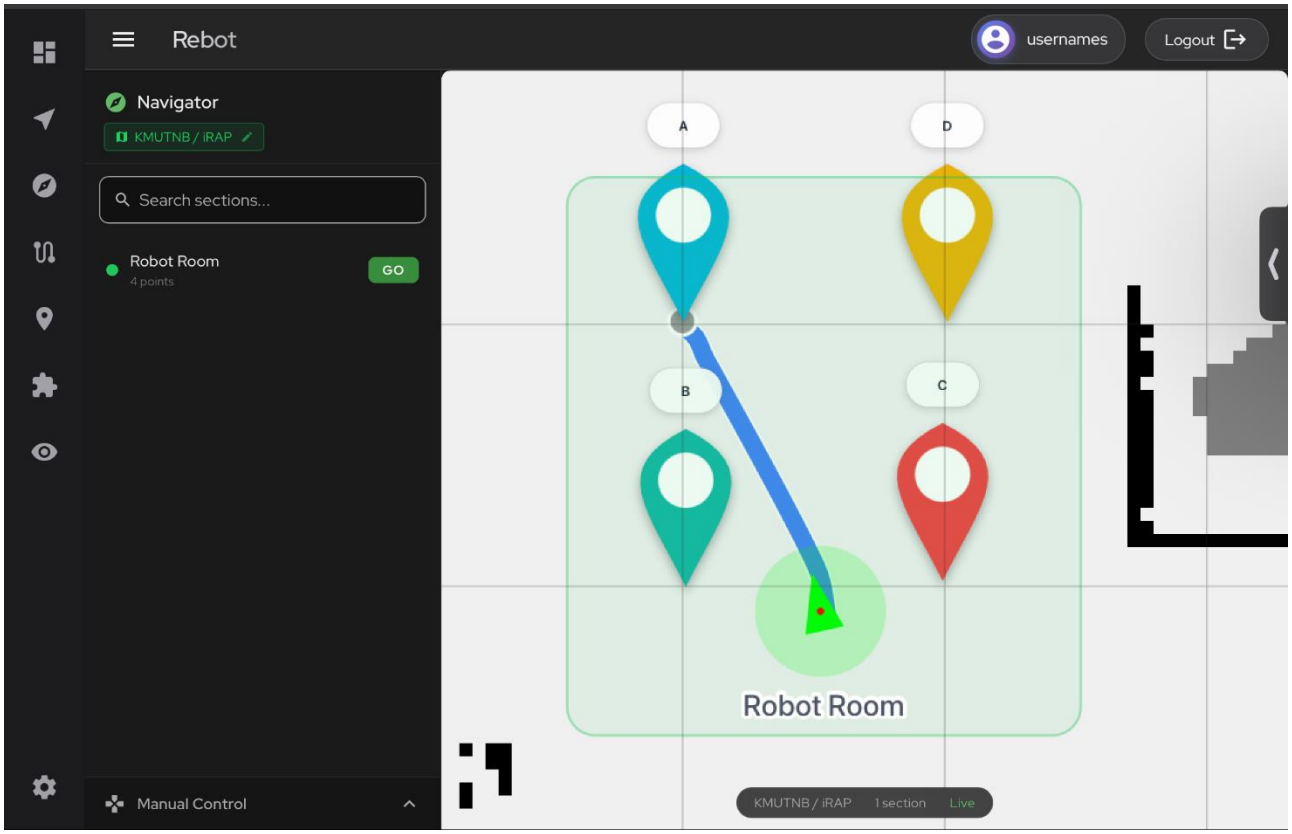


Figure 7-1: Dashboard map view showing robot position, zones, and markers

7.2 Navigation Control

18. Select the target zone from the list on the right panel.
19. Click GO. A dialog shows all Markers within that zone.
20. Select the desired Marker.
21. The system sends a GOAL_COMMAND. The robot plans and executes the path automatically.

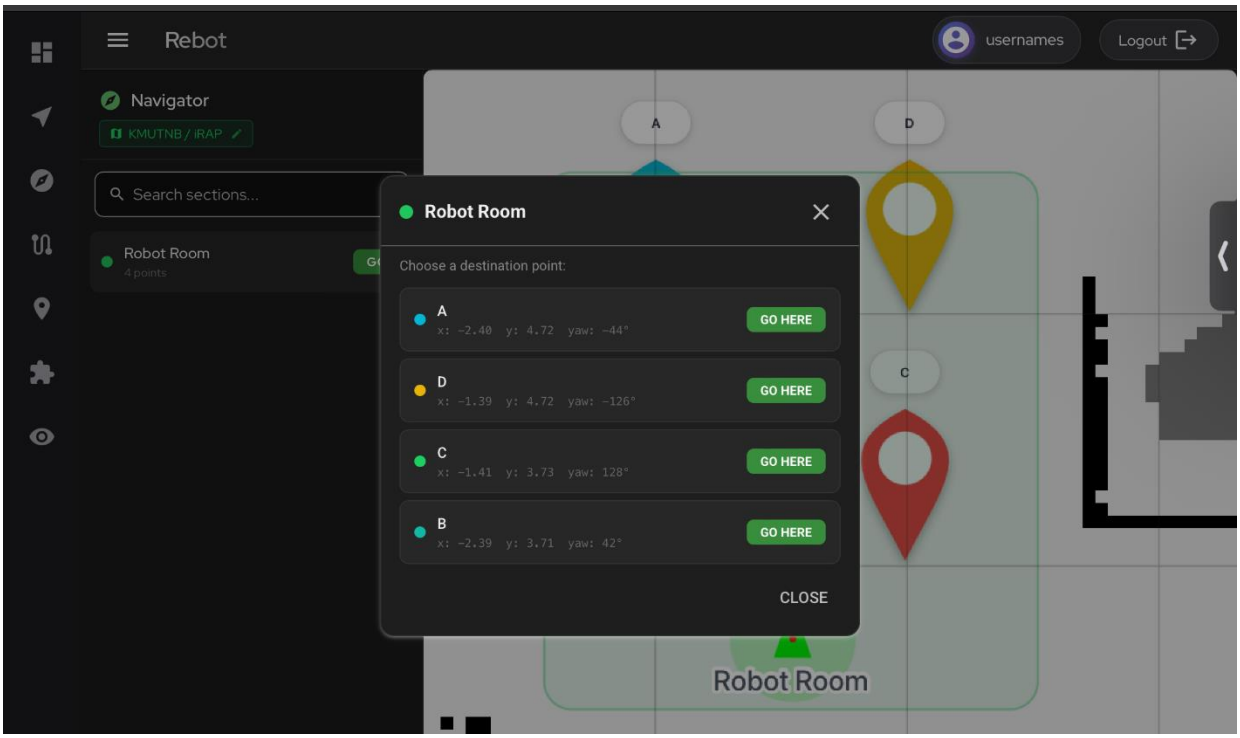


Figure 7-2: Dashboard navigation panel with zone list and GO button

7.3 Routines — Loop Mode

Routines allow the robot to visit multiple Markers in a defined sequence repeatedly. Useful for patrol, inspection, or delivery workflows.

22. Open the Routines panel on the Dashboard.
23. Add Markers in the desired sequence.
24. Enable Loop and click Start.
25. Click Stop at any time to halt.

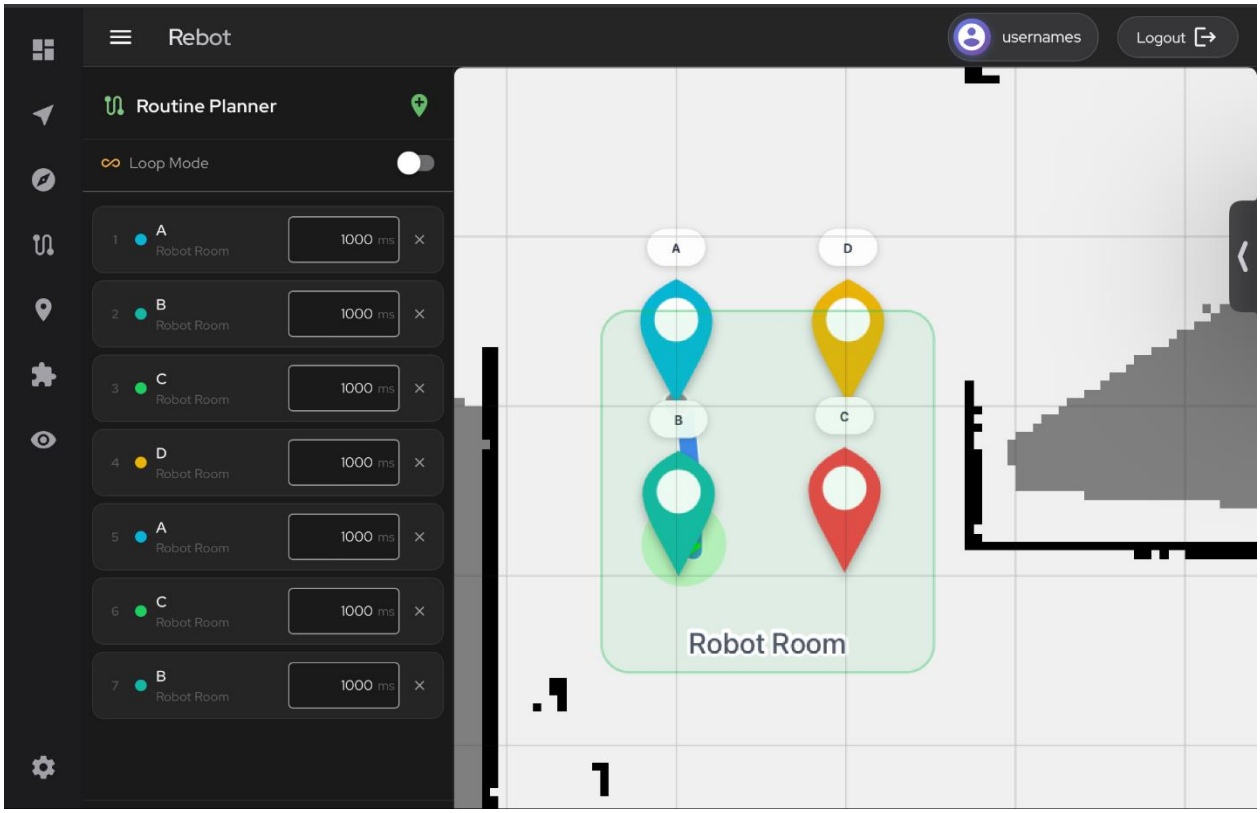


Figure 7-3: Routines panel showing waypoint sequence and Loop toggle

8. Recording Mode — Map Creation

This chapter covers the full process of creating a new map. Maps are built using SLAM while driving the robot through the environment.

8.1 Understanding Sites and Maps

Site	Top-level grouping representing a physical location (e.g., a building floor). All maps that must be linked belong to the same Site.
Map	A 2D OccupancyGrid of a specific area recorded with LiDAR. Multiple maps can exist under one Site.

Example: Create a Site called "CIT", then create separate Maps for each floor or room.

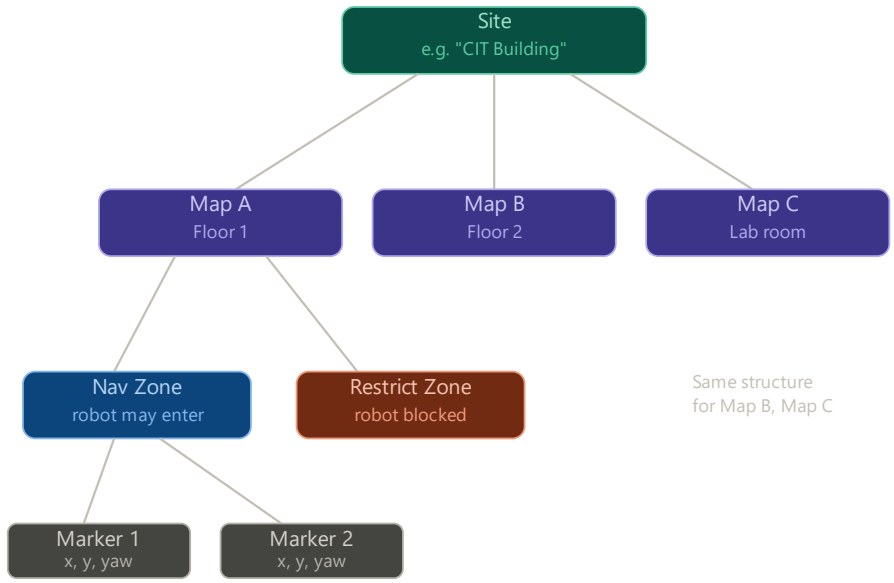


Figure 8-1: Site and Map hierarchy diagram

8.2 Creating a Site

26. Navigate to Setup → Maps.
27. Enter the site name in the "New site name" field.
28. Click CREATE. The Site appears as a green tag.
29. To delete a Site, click the X on its tag.

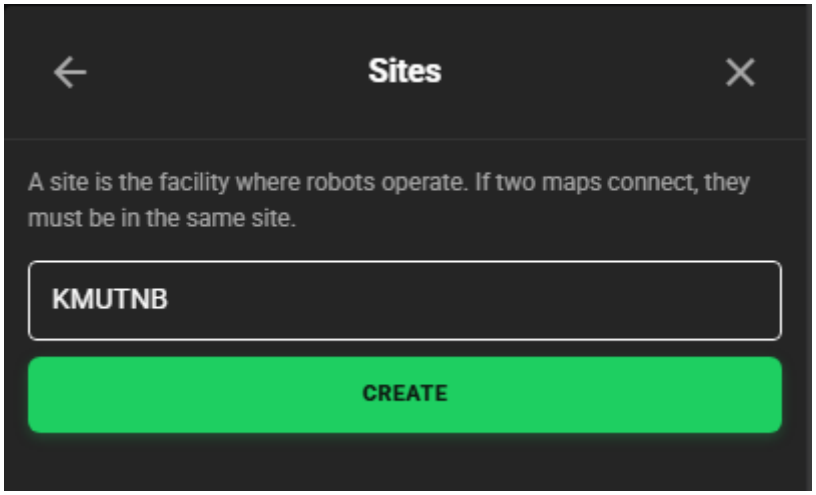


Figure 8-2: Maps page showing the Create Site field and site tags

8.3 Creating a Map Entry

30. Click Create Map on the Maps page.
31. Enter a map name (e.g., "Lab-Floor1").
32. Select the Site from the dropdown. If the Site does not exist, click CREATE / EDIT SITES to add it first.
33. Click CREATE. The new Map entry appears in the list.

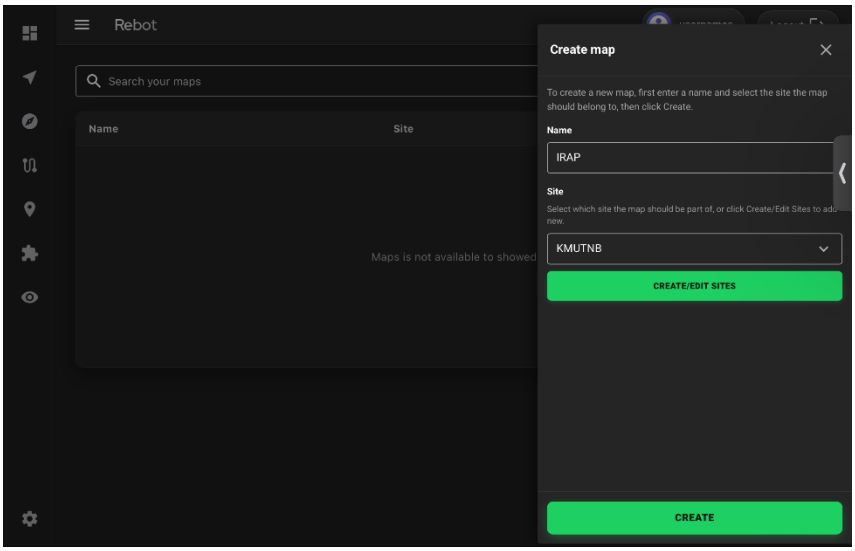


Figure 8-3: Create Map dialog showing the Name field and Site dropdown

8.4 Recording Status Check

Before recording begins, RERIS displays a status check dialog verifying that all required robot systems are active. The following items are checked:

Robot Odometry	Confirms that /odom data is being received. Required for robot localization.
Robot Mapping	Confirms that slam_toolbox is active and publishing /map data.
Robot Connection	Confirms that the WebRTC data channel between the browser and the robot backend is established.

All three items must show a checkmark before recording can proceed. If any item is still loading, wait until it resolves or check the topic configuration in Settings.

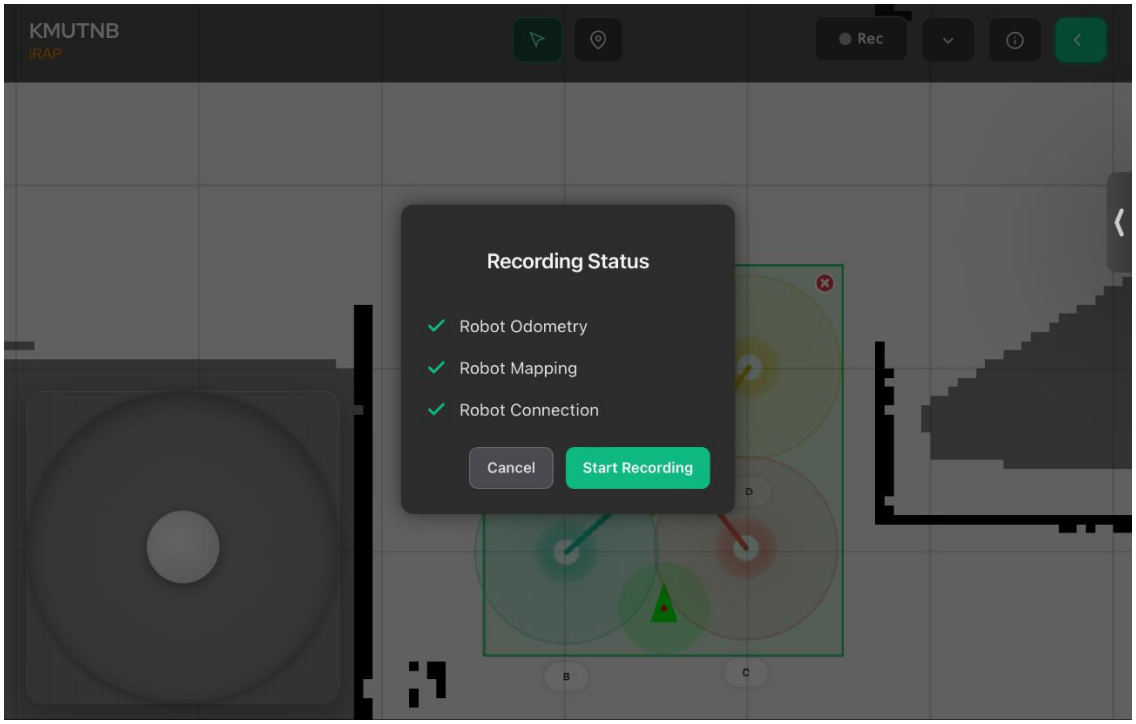


Figure 8-4: Recording Status Check dialog showing Odometry, Mapping, and Connection status

8.5 SLAM Recording Workflow

34. Select the Map entry and click Rec to open the recording status check.
35. Wait for all three status items to confirm, then click Proceed.
36. Use the Teleoperation joystick to drive the robot through all areas to be mapped.
37. The live OccupancyGrid is rendered on screen via WebRTC as the robot scans.
38. When coverage is complete, click the Rec button again to stop. The timer in the Rec button shows elapsed recording time.
39. Proceed to the Toolbox to define Zones and Markers before exporting.

NOTE Drive slowly through doorways and narrow corridors to allow the LiDAR to capture accurate boundary data.

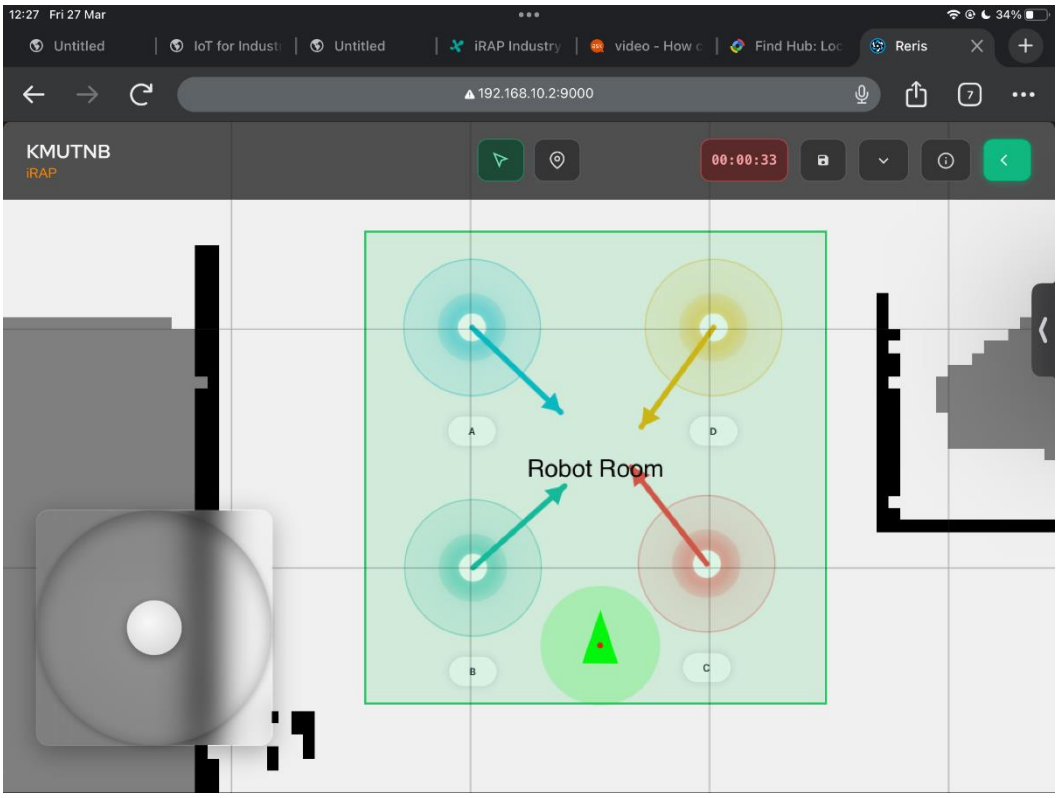


Figure 8-5: Map view during active SLAM recording showing the live-building occupancy grid

9. Toolbox — Map Tools Reference

The Toolbox is the toolbar displayed at the top of the map view during recording and editing. It provides tools for placing goals, drawing zones, and editing the map. Tools are activated by clicking their icon or using the keyboard shortcut shown in the title.



Figure 9-1: Toolbox toolbar showing all available tool buttons

9.1 Tool Overview

Cursor (V)	Default selection and pan tool. Interact with existing markers and zones.
Point-to-Point — Single Goal (P)	Place a single navigation goal on the map.
Multi-Waypoint (Multi Goal)	Place multiple sequential navigation goals.
Area Zone (A)	Draw rectangular zones on the map.
Map Editor — Brush (E)	Paint occupied (black) cells onto the map.
Map Editor — Erase	Remove occupied cells (paint free/white) from the map.
Map Editor — Brush Erase	Circular brush that erases occupied cells. Useful for removing noise patches.
Rec / Timer	Start or stop the SLAM recording session. Shows elapsed time while recording.
Save Draft	Save current Zones and Markers without ending the recording session.
Inspection Toggle	Open or close the Tool Inspection Panel (see Chapter 10).

9.2 Cursor Tool (V)

The Cursor is the default tool and is active whenever no other tool is selected. It provides the following functions:

- Pan the map view by clicking and dragging on empty canvas area.
- Select an existing Single Goal Marker by clicking on it — the Inspection Panel updates to show its properties.
- Select an existing Multi-Goal Marker or Area Zone by clicking — enables editing in the Inspection Panel.
- Interact with UI elements overlaid on the map (zone labels, marker pins).

9.3 Point-to-Point Tool — Single Goal (P)

This tool places a single navigation goal on the map. Only one Single Goal can exist at a time.

40. Select the PTP tool (P).
41. Click on the map at the desired destination. A pin appears showing the position (x, y).
42. Hold and drag to set the heading (Yaw) direction. A preview arrow shows the target heading.
43. Release to confirm. The goal is saved in the Inspection Panel under Single Marker.
44. When ready, send the goal to the robot from the Inspection Panel or Dashboard.

> **NOTE** The coordinates are in the ROS 2 world frame (metres). Yaw is displayed in degrees in the Inspection Panel but stored in radians internally.

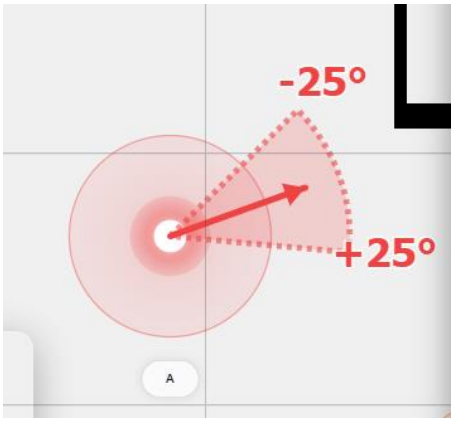


Figure 9-3: PTP tool placing a single goal with heading arrow visible

9.4 Multi-Waypoint Tool (Multi Goal)

The Multi-Waypoint tool adds sequential navigation goals. Each waypoint is a separate Marker with its own position, heading, tolerance, and delay settings.

45. Select the Multi-Waypoint tool.
46. Click on the map to place the first waypoint.
47. Drag to set the heading, then release to confirm.
48. Repeat for each subsequent waypoint. Markers are numbered in order of placement.
49. All waypoints are visible and editable in the Inspection Panel under Multi Markers.

Each Multi Marker supports the following configurable action parameters (editable in the Inspection Panel):

Radius (m)	Arrival tolerance radius. The robot considers the goal reached when it enters this radius. Range: 0.1–5.0 m.

Yaw Error (°)	Acceptable heading error at arrival. Range: 1–180°. Lower values require more precise heading alignment.
Delay (ms)	Wait time at this waypoint before proceeding to the next. Useful for tasks requiring the robot to pause at a location.

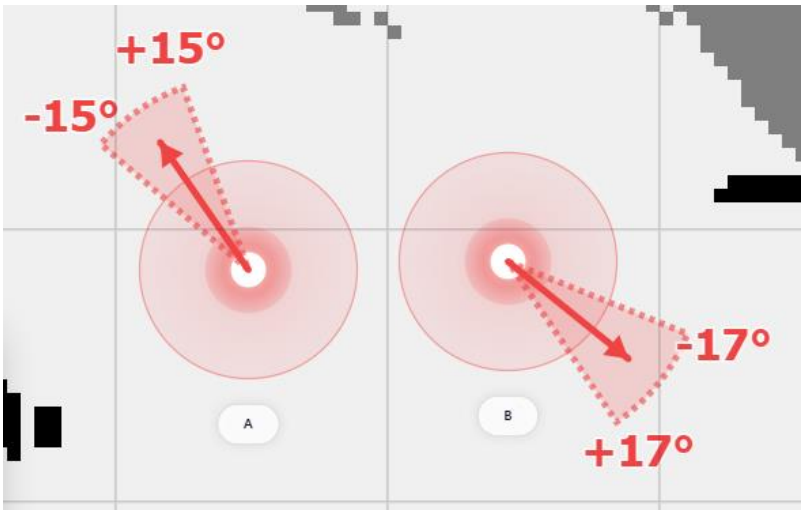


Figure 9-4: Multi-Waypoint tool placing sequential markers with numbered labels

9.5 Area Zone Tool (A)

The Area Zone tool draws rectangular zones on the map. Zones group Markers and define operational or restricted regions.

50. Select the Area tool (A).
51. Click and drag on the map to draw a rectangle.
52. Release to confirm. The zone appears in the Inspection Panel under Areas.
53. In the Inspection Panel, assign a label, colour, and type (Navigation or Restrict).

Navigation Zone	Operational area the robot may navigate within. Displayed in the assigned colour.
Restrict Zone	Forbidden area. Displayed in red. Nav2 will not plan paths through this zone.



Figure 9-5: Area Zone tool drawing a rectangle and the zone type toggle in the Inspection Panel

9.6 Map Editor — Brush (E)

The Brush tool paints occupied cells (black/obstacle) directly onto the OccupancyGrid. Use it to add walls, close doorways, or mark areas that should be treated as obstacles.

- Click and drag to paint. The brush traces a continuous filled stroke.
- Adjust brush radius in the toolbar — available in metres (converted to pixels using map resolution) or direct pixel units.
- Nav2 will treat all painted cells as obstacles and avoid them when planning paths.

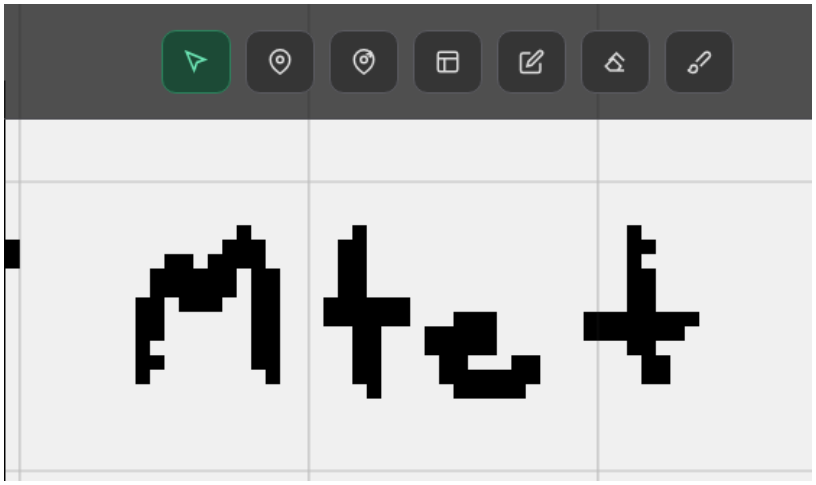


Figure 9-6: Brush tool painting occupied cells onto the map

9.7 Map Editor — Erase

The Erase tool removes occupied cells by painting free (white) cells onto the map. Use it to delete incorrectly captured obstacles or sensor noise that blocks valid paths.

- Click and drag to erase. The tool traces a continuous stroke removing occupied cells.
- Adjustable brush radius as with the Brush tool.
- Erased cells become free space — Nav2 will allow path planning through them.

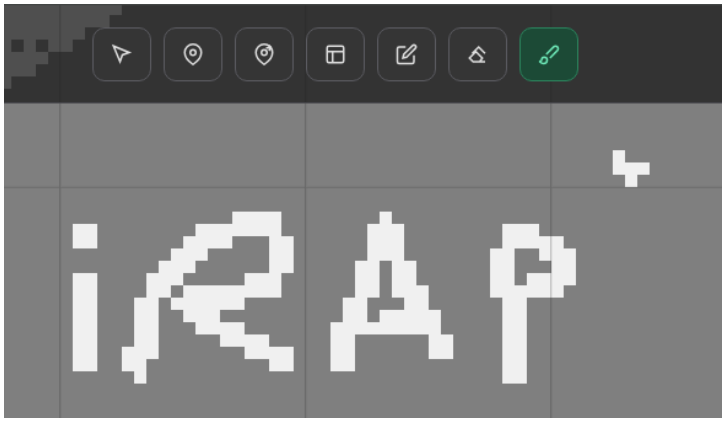


Figure 9-7: Erase tool removing noise from the map

9.8 Map Editor — Brush Erase

The Brush Erase is a circular variant of the Erase tool. It is particularly effective for removing large patches of sensor noise or clearing circular obstacle artefacts from the map.

- Operates as a circular brush rather than a line stroke.
- All cells within the brush radius are set to free on each click or drag step.
- Brush radius adjustable in the same manner as the standard Brush and Erase tools.

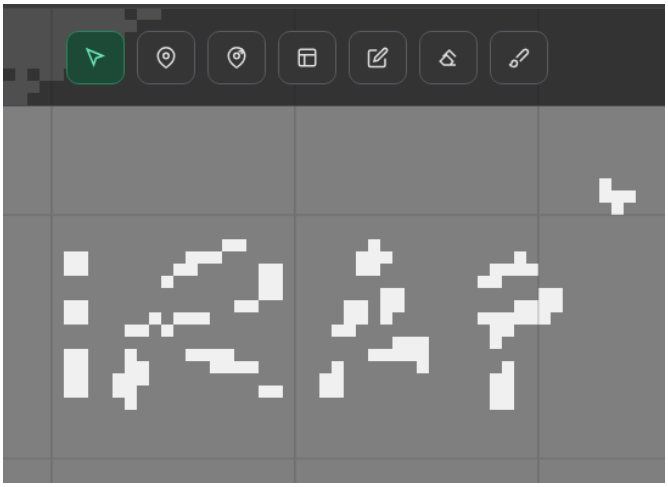


Figure 9-8: Brush Erase tool removing a circular patch of noise

9.9 Rec Button and Recording Timer

The Rec button in the top-right of the Toolbox starts and stops the SLAM recording session:

- Before recording: button is labelled Rec with a pulsing dot indicator.
- While recording: the button displays a live timer (HH:MM:SS) counting elapsed recording time.
- While saving: the button is disabled and displays Saving until the process completes.
- Click the button again while recording to stop the session and trigger the save process.

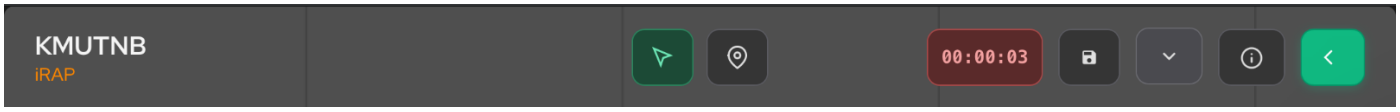


Figure 9-9: Rec button in idle state, active recording state, and saving state

9.10 Save Draft

The Save Draft button (floppy disk icon) saves the current state of all Zones and Markers to the server without ending the recording session. Use this to checkpoint progress during long mapping sessions in case of interruptions.

> **NOTE** Save Draft does not export the OccupancyGrid map data. To save the full map, use the Export function after the recording session ends.

10. Tool Inspection Panel

The Tool Inspection Panel (Layers panel) is a collapsible side panel that displays and allows editing of all map objects placed on the canvas — Single Markers, Multi Markers, and Area Zones. It is the primary interface for fine-tuning positions, headings, labels, colours, and action parameters.

10.1 Opening the Panel

Click the Inspection Toggle button (circle-i icon) in the top-right of the Toolbox to open or close the panel. The panel can also be collapsed by clicking its header bar.

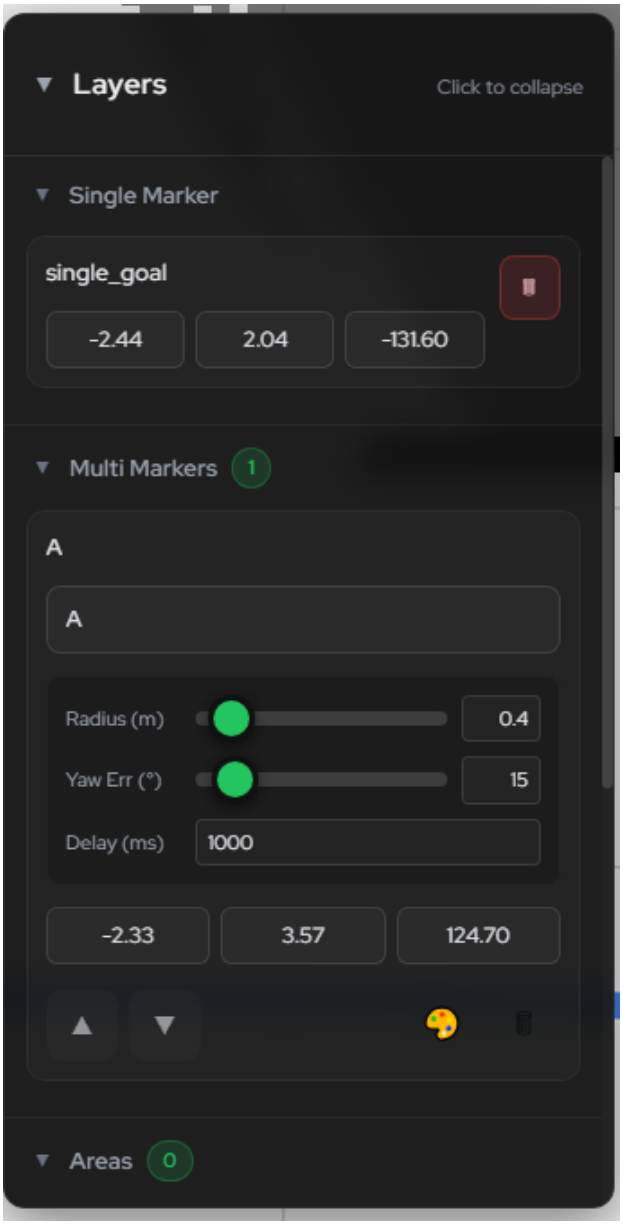


Figure 10-1: Tool Inspection Panel open showing all three sections: Single Marker, Multi Markers, and Areas

10.2 Single Marker Section

This section appears only when a Single Goal has been placed on the map. It displays the current position and heading of the Single Goal Marker and allows direct editing.

X	X coordinate of the goal in the ROS 2 world frame (metres). Directly editable — press Enter or click away to confirm.
Y	Y coordinate of the goal in the ROS 2 world frame (metres). Directly editable.
Yaw	Heading of the robot at arrival, displayed in degrees. Directly editable. Internally stored in radians.
Delete button	Removes the Single Goal from the map. The section disappears until a new Single Goal is placed.

NOTE All position fields support draft editing — changes take effect only when you press Enter or click away (blur). Press Escape to revert a draft without applying.

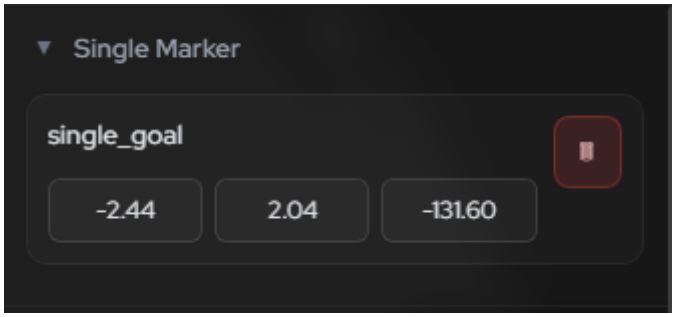


Figure 10-2: Single Marker section in the Inspection Panel showing X, Y, Yaw fields and delete button

10.3 Multi Markers Section

This section lists all Multi-Goal Markers in the order they will be visited by the robot. Each Marker row is fully editable.

Label	Display name for the waypoint. Editable inline. Used in the Dashboard navigation list.
Radius (m)	Arrival tolerance radius — slider and number input. Range: 0.1–5.0 m.
Yaw Error (°)	Acceptable heading error at arrival — slider and number input. Range: 1–180°.
Delay (ms)	Wait time at this waypoint before moving to the next. Number input.
X / Y / Yaw	World-frame position and heading in degrees. All directly editable.
Colour picker	Click the palette icon to open a colour swatch panel and assign a display colour.
Reorder (▲ / ▼)	Move the marker up or down in the execution sequence.
Executing badge	A green ▶ Executing badge appears on the Marker currently being targeted by the robot.
Delete button	Removes the Marker from the sequence.

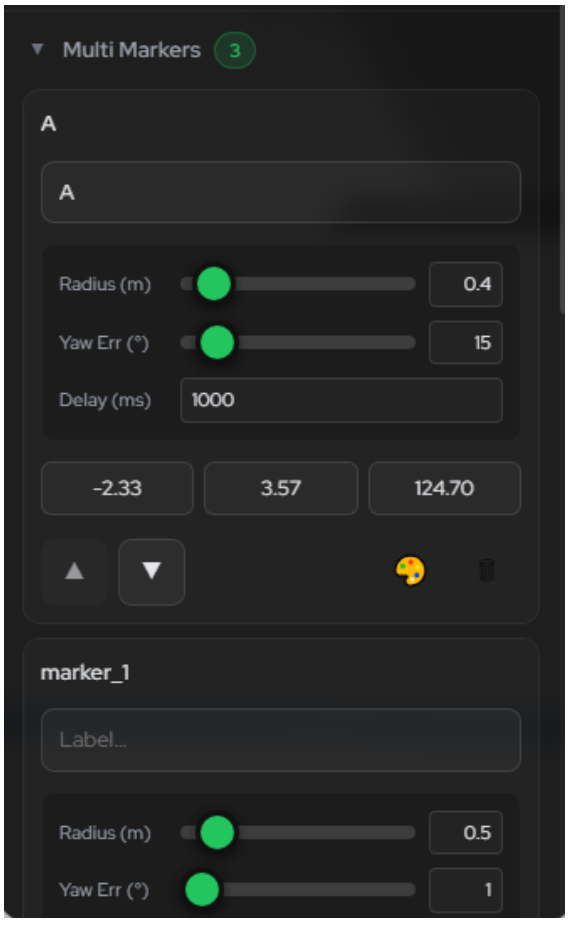


Figure 10-3: Multi Markers section showing several waypoints with action parameters, reorder buttons, and the colour picker

10.4 Areas Section

This section lists all Area Zones drawn on the map. Each Area can be renamed, retyped, and recoloured.

Label	Display name for the zone. Editable inline.
Zone Type Toggle	Switches between Navigation (blue) and Restrict (red) zone types.
Colour picker	Assigns a display colour to the zone boundary.
Centre coordinates	Read-only display of the zone's centre position in world coordinates.
Delete button	Removes the Zone from the map.

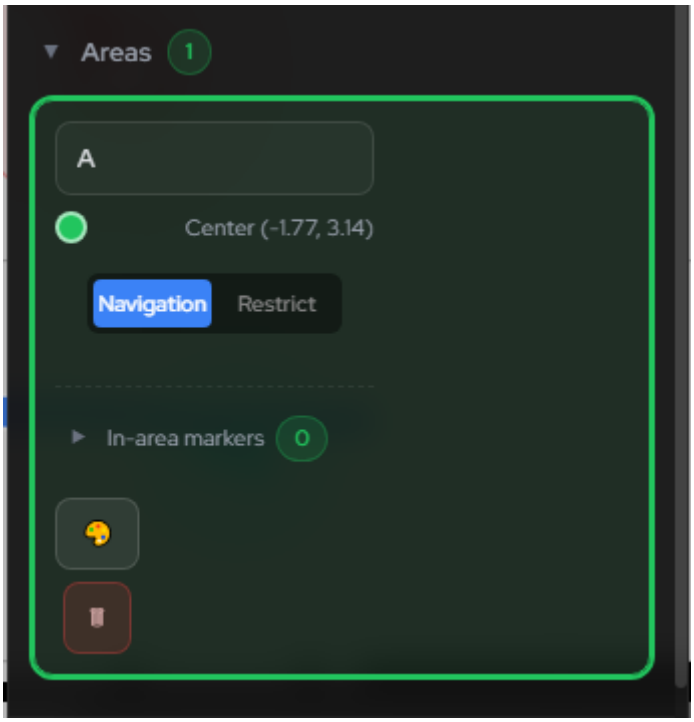


Figure 10-4: Areas section showing a zone with label, type toggle, colour picker, and in-area markers

10.5 In-area Markers (Dock Markers)

Within each Area section, an expandable sub-list shows all Markers that are currently positioned inside that Zone's boundaries. These are called Dock Markers. They are the same Multi-Goal Markers that appear in the Multi Markers section — the system automatically detects which markers fall within a zone's rectangle.

- Dock Markers can be renamed directly within the Area's sub-list.
- Colour can be changed via the palette icon.
- When the user sends a navigation command from the Dashboard, they select a Zone and then a Dock Marker within it. The system sends the Marker's (x, y, yaw) as the goal to Nav2.

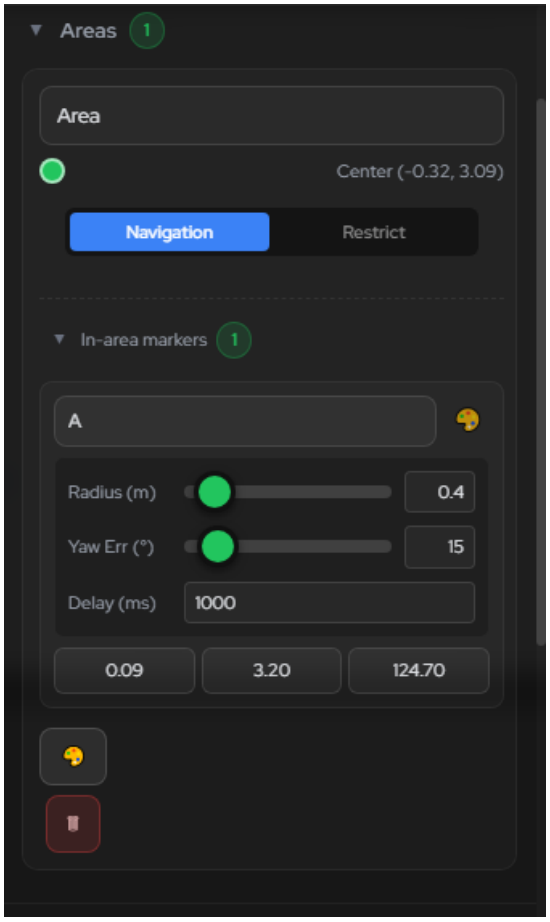


Figure 10-5: In-area Markers sub-list expanded inside an Area row showing dock marker labels and controls

11. Saving the Map — Export

After recording and annotation are complete, export the map to permanently save all data to the robot's database.

11.1 Export Workflow

54. Ensure all Zones and Markers are correctly placed and labelled in the Inspection Panel.
55. Click Export Map in the Toolbox area.
56. The system compiles the OccupancyGrid, all Area Zones, and all Marker positions (including action parameters).
57. Data is sent to the RERIS backend and stored in MongoDB on the robot.
58. The map is immediately available in the Dashboard for navigation without restarting.

> **NOTE** Exporting overwrites any previously exported version of this map. Use Save Draft during recording to preserve intermediate states.

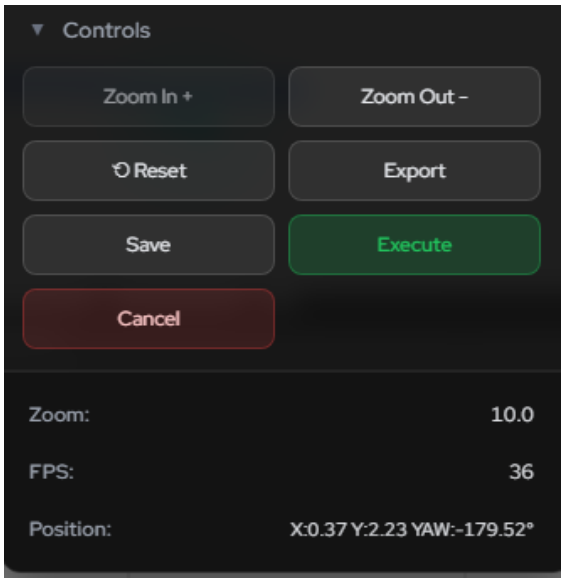


Figure 11-1: Using Save button to export to currently devices

12. Map Editor — Advanced Editing

The Map Editor allows post-recording correction of the OccupancyGrid. Access it via Setup → Maps → Modify after a map has been recorded.

12.1 Editor Tools Reference

Brush Tool (E)	Paints occupied cells (black). Adds walls or obstacles not captured during recording.
Erase Tool	Paints free cells (white). Removes noise that incorrectly blocks navigable paths.
Brush Erase Tool	Circular erase brush for efficient removal of large noise patches.
Brush Radius	Adjustable in metres or pixels. Converted using the map's resolution parameter.

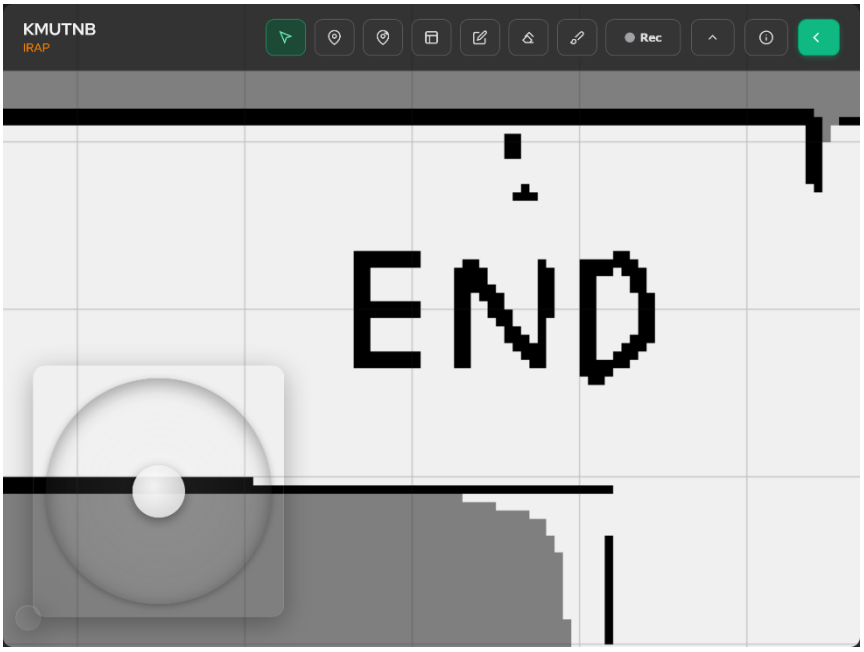


Figure 12-1: Map Editor view with Brush tool active showing a painted obstacle on the map

12.2 Undo and Redo

The Map Editor maintains a full Undo/Redo history at the stroke level. Each continuous drag is recorded as a single undoable action.

Undo	Ctrl + Z — reverts the most recent stroke.
Redo	Ctrl + Y — reapplies a reverted stroke.

NOTE History is stored in memory only. Navigating away from the Map Editor without exporting will discard all unsaved changes.

13. Profile

13.1 Account

View and update your display name and account details via Profile → Account.

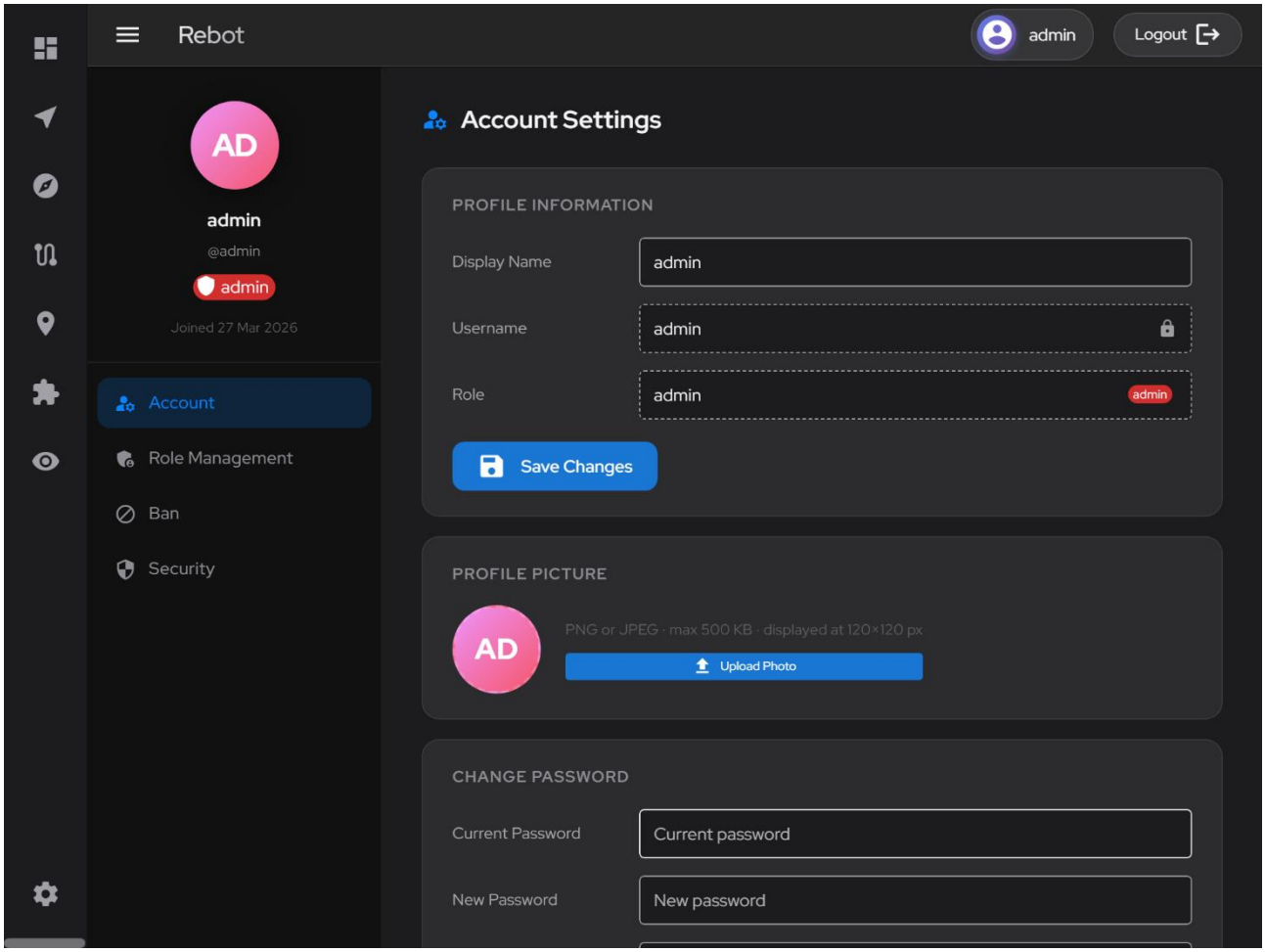


Figure 13-1: Profile — Account page

13.2 Security

Change your password via Profile → Security. You must provide your current password and your Security Code to confirm the change.

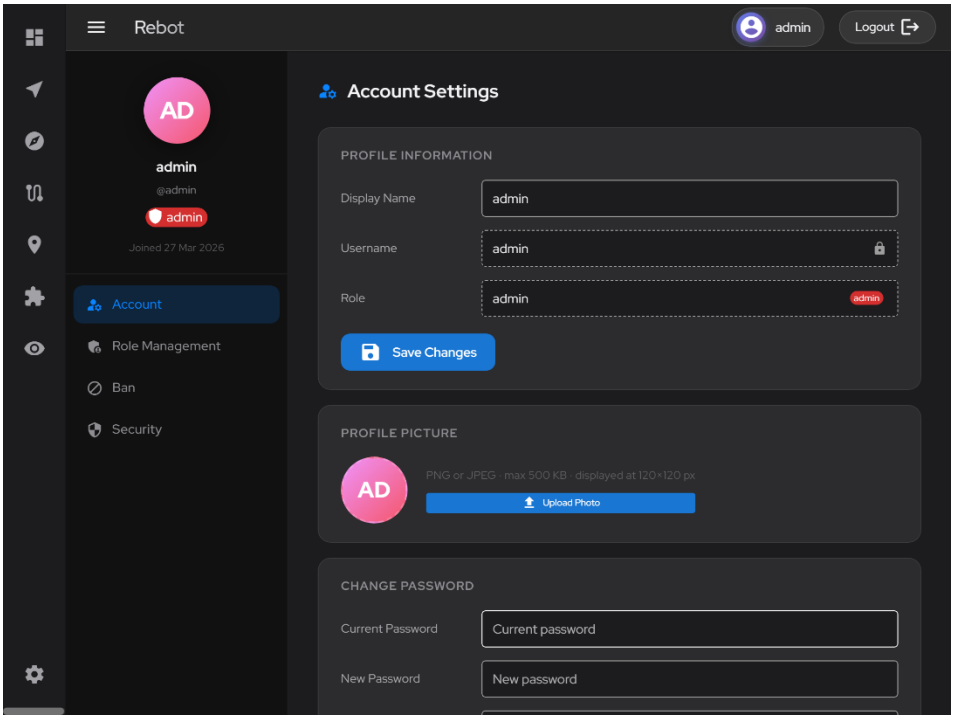


Figure 13-2: Profile — Security page showing the password change form

14. Glossary

RERIS	Robot Interface System. The web-based control interface
ROS 2	Robot Operating System 2. The middleware framework for the robot's hardware and navigation stack.
SLAM	Simultaneous Localisation and Mapping. The robot builds a map while tracking its own position.
slam_toolbox	ROS 2 package providing SLAM capability. Publishes the OccupancyGrid map.
Nav2	ROS 2 Navigation Stack. Handles path planning, obstacle avoidance, and autonomous movement.
/cmd_vel	ROS 2 topic for velocity commands (Vx linear, Vy lateral). Published by RERIS to control the robot.
/odom	ROS 2 topic for odometry data (position + orientation). Published by the robot's hardware driver.
/scan	ROS 2 topic for LiDAR scan data. Consumed by slam_toolbox to build the map.
/imu	ROS 2 topic for IMU sensor data. Used for heading fusion in slam_toolbox.
WebRTC	Web Real-Time Communication. Peer-to-peer data streaming between browser and robot.
OccupancyGrid	2D grid map format. Each cell: Free (0), Occupied (100), Unknown (-1).
Site	Top-level grouping of all maps belonging to one physical location.
Zone	Named rectangular area on a map. Can be Navigation or Restrict type.
Marker	Waypoint with position (x, y) and heading (yaw). Used as navigation targets.
Dock Marker	A Marker that falls within a Zone's boundary. Shown in the In-area Markers list.
Single Goal	A one-time navigation target. Only one can exist at a time.
Multi Goal	An ordered sequence of waypoints. Supports Radius, Yaw Error, and Delay per point.
Security Code	6-digit code generated at registration. Required for offline password reset.
JWT	JSON Web Token. Secure token issued on login. Required for all authenticated API calls.
MongoDB	NoSQL document database. Stores user accounts, profiles, and map data.
colcon	ROS 2 build tool for compiling workspaces.
systemd	Linux service manager used to run RERIS as an auto-start service.